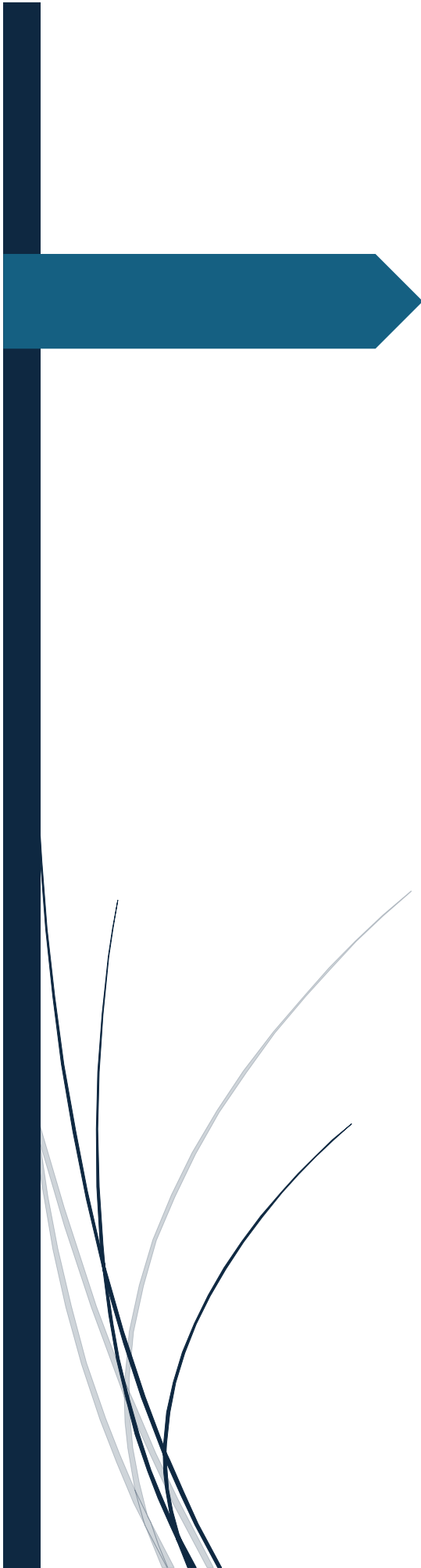




FreshRetailNet-50K

Final Report

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FreshRetailNet-50K: Demand Forecasting, Stockout Risk & Inventory Intelligence

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1. Executive Summary

FreshRetailNet-50K: Demand Forecasting, Stockout Risk & Inventory Intelligence is an end-to-end analytics and forecasting project designed to support demand planning, inventory monitoring, and operational decision-making in the fresh retail sector. The project focuses on understanding product-store demand behavior, identifying stockout exposure, evaluating the effect of discounts and external conditions, and building forecasting models that can support future inventory and replenishment decisions.

Fresh retail businesses face a critical operational challenge: demand is often volatile, products are perishable, and inaccurate inventory planning can lead to either stockouts, lost sales, and poor customer availability, or overstocking, waste, and unnecessary operational cost. This makes demand forecasting and inventory risk monitoring especially important for products with short shelf lives and changing demand patterns.

The project uses the **FreshRetailNet-50K** dataset, which provides detailed store-product time series data for fresh retail products. The dataset includes sales information, store and product identifiers, stockout-related indicators, discount levels, holiday and activity flags, and weather-related variables. These features make the dataset suitable for both business intelligence analysis and predictive modeling.

The analytical workflow combines **SQL, Python, and Power BI**. SQL is used to structure the data, validate the analytical grain, perform data quality checks, and prepare business-ready analytical views. Python is used for data understanding, exploratory data analysis, feature engineering, forecasting model development, backtesting, and model evaluation. Power BI is used to deliver a business-facing dashboard that communicates demand trends, stockout patterns, promotional impact, and forecasting results in an accessible decision-support format.

From a **Data Analyst** perspective, the project investigates demand trends, seasonality, store and product performance, category contribution, discount behavior, holiday effects, weather-related demand variation, and stockout patterns. These analyses help identify which products and locations drive demand, where operational risk is concentrated, and which factors influence sales behavior.

From a **Data Scientist** perspective, the project develops a forecasting framework using baseline and machine learning approaches. The modeling phase starts with simple benchmark methods such as naive forecasting, moving averages, and seasonal naive models, then progresses toward machine learning models using lag features, rolling statistics, calendar variables, stockout indicators, discount features, and weather inputs. Forecasting performance is evaluated through backtesting using metrics such as MAE, RMSE, WAPE, and forecast bias.

The final output is not only a forecasting exercise, but a broader **Inventory Risk Intelligence Framework**. Forecasted demand, recent stockout history, demand volatility, promotional sensitivity, and product-store behavior are combined to classify products and stores into actionable inventory risk segments. These segments can support replenishment prioritization, stockout monitoring, promo planning, and operational review.

Overall, this project demonstrates the ability to build a complete analytics solution that connects business context, data preparation, exploratory analysis, feature engineering, forecasting, evaluation, dashboarding, and business recommendations. It is designed to showcase practical skills across both **Data Analysis** and **Data Science**, with a strong focus on translating technical outputs into business decisions.

2. Business Context

Fresh retail is a highly time-sensitive retail environment where products are often perishable, demand can change quickly, and inventory decisions directly affect both sales performance and operational efficiency. Unlike standard retail categories, fresh products usually have shorter shelf lives, higher spoilage risk, and more frequent demand fluctuations. As a result, businesses operating in this sector must carefully balance product availability with inventory waste.

One of the main challenges in fresh retail is maintaining the right stock level at the right store and time. If inventory is too low, customers may face product unavailability, leading to stockouts, missing sales opportunities, and reduced customer satisfaction. If inventory is too high, the business may experience overstocking, product waste, markdown pressure, and unnecessary holding costs. This creates a strong need for accurate demand forecasting and initiative-taking inventory risk monitoring.

Demand in fresh retail is influenced by several factors, including store location, product category, day-of-week patterns, holidays, promotional activity, discounts, and weather conditions. These factors can create different demand behaviors across stores and products, making simple historical sales reporting insufficient for effective planning. A more structured analytical approach is needed to understand what drives demand, which products are most volatile, and where inventory risk is most concentrated.

Stockouts are especially important in this context because observed sales may not always represent true customer demand. When a product is out of stock, sales may appear low even though actual customer demand could have been higher. This creates a censored demand problem, where the business may underestimate future demand if stockout behavior is not considered during analysis and forecasting.

This project addresses these business challenges by building a demand forecasting and inventory intelligence framework using store-product time series data. The goal is to help business users understand historical demand patterns, identify stockout exposure, evaluate the impact of discounts and external factors, and generate forecasting outputs that can support better replenishment and inventory planning decisions.

By combining SQL, Python, and Power BI, the project connects technical analysis with business decision-making. SQL supports structured data preparation and validation, Python supports exploratory analysis, feature engineering, forecasting, and model evaluation, while Power BI provides a business-facing reporting layer for monitoring demand trends, stockout risk, and forecasting performance.

Overall, the business context of this project is centered on improving fresh retail decision-making by transforming raw sales, stockout, promotion, and weather data into actionable demand and inventory insights. The project is designed to support more informed replenishment decisions, reduce the risk of stockouts and overstocking, and improve the company's ability to plan inventory based on data-driven forecasts rather than reactive historical reporting alone.

3. Business Problem

Fresh retail businesses operate in an environment where demand is highly variable, products are perishable, and inventory decisions must be made frequently at a granular store-product level. In this context, relying only on historical sales reporting can lead to inaccurate planning because recorded sales do not always reflect true customer demand, especially when products experience stockouts.

A key business problem is that low sales may be caused either by weak demand or by product unavailability. When a product is out of stock, observed sales are artificially limited, which can cause the business to underestimate future demand. This creates a censored demand problem: the company may interpret low sales as low customer interest, when in reality customers may have wanted to buy the product but could not because inventory was unavailable.

At the same time, overestimating demand can also create operational and financial issues. Fresh products often have limited shelf life, so excess inventory can lead to waste, markdowns, spoilage, and unnecessary holding costs. This makes demand planning more complex than simply maximizing stock availability. The business must balance two competing risks: avoiding stockouts while also preventing overstock.

Demand patterns are also affected by multiple external and operational factors, including discounts, activity periods, holidays, store location, product category, and weather conditions. Without a structured analytical framework, it becomes difficult to understand which factors are driving demand changes, which products are most sensitive to promotions, and which stores or products require closer inventory monitoring.

The core business problem is therefore the absence of an integrated demand and inventory intelligence framework that connects historical sales, stockout behavior, promotional activity, weather conditions, and forecasting outputs into one decision-support process. Without this framework, inventory decisions may remain reactive, based mainly on past sales rather than forward-looking demand signals and operational risk indicators.

This project addresses the problem by developing a structured approach to analyze demand behavior, detect stockout exposure, evaluate business drivers of sales variation, forecast future demand, and translate those forecasts into actionable inventory risk segments. The goal is to help fresh retail decision-makers improve replenishment planning, reduce missed sales from stockouts, limit overstock risk, and support more data-driven inventory decisions.

4. Project Objectives

The main objective of this project is to develop an end-to-end Demand Forecasting, Stockout Risk, and Inventory Intelligence Framework for fresh retail operations. The project aims to transform raw store-product time series data into actionable business insights, forecasting outputs, and inventory risk indicators that can support more informed replenishment and planning decisions.

The project is designed to achieve the following objectives:

1. **Understand historical demand behavior**

Analyze sales patterns across stores, products, product categories, cities, and dates in order to identify key demand trends, seasonality, and demand concentration.

2. **Assess product and store performance**

Evaluate which products, stores, and product categories contribute most to sales activity, and identify high-performing, low-performing, stable, and volatile demand segments.

3. **Analyze stockout exposure**

Examine stockout indicators to identify where product availability issues are concentrated and how stockouts may affect observed sales behavior.

4. **Evaluate the impact of business and external factors**

Measure how discounts, activity flags, holidays, and weather-related variables are associated with changes in sales demand across various products and stores.

5. **Create a forecasting-ready feature layer**

Build structured features such as lag variables, rolling statistics, calendar indicators, stockout features, discount variables, and weather inputs to support predictive modeling.

6. **Develop and compare forecasting models**

Build baseline forecasting models such as naive forecasts, moving averages, and seasonal naive models, then compare them with more advanced machine learning models such as XGBoost or LightGBM.

7. **Evaluate forecasting performance through backtesting**

Use time-based validation methods to assess model performance using metrics such as MAE, RMSE, WAPE, and forecast bias.

8. **Build an inventory risk framework**

Combine forecasted demand, recent sales behavior, stockout history, demand volatility, and promotional sensitivity to classify products and stores into actionable inventory risk groups.

9. **Deliver a business-facing Power BI dashboard**

Create an interactive dashboard that allows business users to monitor demand trends, product/store performance, stockout risk, discount, and weather impact, and forecasting performance.

10. **Translate analytical findings into business recommendations**

Provide practical recommendations to support replenishment prioritization, stockout monitoring, promotion planning, and more data-driven inventory decision-making.

Overall, the project aims to demonstrate how SQL, Python, and Power BI can be combined to build a complete analytics solution that connects data preparation, business analysis, forecasting, inventory risk assessment, and executive reporting into one integrated workflow.

5. Stakeholders / Business Users

This project is designed to support multiple business and technical stakeholders involved in fresh retail demand planning, inventory management, and operational decision-making. Because the project connects historical sales behavior, stockout exposure, promotional activity, weather conditions, forecasting outputs, and inventory risk segmentation, the final analysis can be used by both technical teams and business users.

Executive Leadership

Executive leadership can use the project to understand overall demand performance, inventory risk exposure, and the business impact of stockouts and overstocking. The dashboard and final insights provide a high-level view of demand stability, product availability issues, and the effectiveness of forecasting-driven planning. This helps leadership evaluate whether inventory decisions are being managed proactively or reactively.

Supply Chain and Inventory Planning Teams

Supply chain and inventory planning teams are among the primary users of this project. They can use the analysis to identify products and stores with high stockout risk, monitor demand volatility, prioritize replenishment decisions, and detect potential overstock situations. The inventory risk framework can help these teams move from static replenishment rules toward more data-driven planning based on recent demand behavior and forecasted demand.

Store Operations Teams

Store operations teams can use the project to understand demand patterns at the store level and identify which products require closer monitoring. The analysis can help store managers detect recurring stock availability issues, understand daily and weekly demand cycles, and respond more effectively to local demand changes. This is especially important for fresh products where availability and timing directly affect customer experience.

Category and Product Managers

Category and product managers can use the project to evaluate product and category performance across stores, cities, and time periods. The analysis helps identify high-demand products, slow-moving products, volatile products, and products that are sensitive to discounts or activity periods. These insights can support assortment planning, category prioritization, promotional planning, and product-level inventory strategies.

Pricing and Promotion Teams

Pricing and promotion teams can use the project to evaluate how discounts, activity flags, holidays, and external conditions are associated with sales behavior. By comparing demand under different discount levels and activity periods, these teams can better understand which products respond strongly to promotional actions and which products show limited uplift. This can support more targeted and efficient promotion planning.

Data and Analytics Teams

Data and analytics teams can use the project as a reusable analytical framework for demand forecasting, feature engineering, backtesting, and business reporting. The SQL preparation layer, Python modeling workflow, and Power BI dashboard structure provide a foundation that can be extended into future forecasting models, automated reporting pipelines, or more advanced inventory optimization solutions.

Finance and Commercial Planning Teams

Finance and commercial planning teams can use the outputs to understand the financial implications of demand variability, stockout exposure, and overstock risk. Although the dataset focuses on normalized sales amounts rather than direct revenue or margin values, the framework still supports business planning by identifying demand concentration, operational risk, and products that may require closer commercial review.

Customer Experience Teams

Customer experience teams can benefit indirectly from the project because product availability is a key part of the customer retail experience. Repeated stockouts may lead to missed purchases and reduced customer satisfaction. By identifying where stockout risk is concentrated, the project can help the business improve product availability and reduce customer-facing availability issues.

Overall, the project is relevant to stakeholders across leadership, supply chain, store operations, category management, pricing, analytics, finance, and customer experience. Its value lies in transforming raw demand, stockout, discount, and contextual data into a decision-support framework that can help different teams make more informed planning and operational decisions.

6. Dataset Source & License

This project uses the **FreshRetailNet-50K** dataset, an open benchmark dataset designed for demand forecasting and inventory-related analysis in the fresh retail sector. The dataset is published by **Dingdong-Inc** on Hugging Face and is intended for time series forecasting research and practical retail demand modeling use cases.

The dataset contains detailed store-product time series records for fresh retail products, including sales behavior, stockout indicators, discount information, holiday and activity flags, and weather-related contextual variables. These features make the dataset highly suitable for analyzing demand trends, identifying stockout exposure, evaluating promotion and discount effects, and building forecasting models for inventory planning.

The official dataset repository provides the data in **Parquet format**, with two main files:

- data/train.parquet
- data/eval.parquet

According to the dataset documentation, FreshRetailNet-50K contains store-product time series data with fields such as city_id, store_id, product_id, dt, sale_amount, hours_sale, stock_hour6_22_cnt, hours_stock_status, discount, holiday_flag, activity_flag, precipitation, temperature, humidity, and wind-level variables. These variables support both daily demand analysis and more granular stockout-aware forecasting workflows.

The dataset is licensed under the **Creative Commons Attribution 4.0 International License (CC BY 4.0)**. This license allows the material to be shared and adapted, including for commercial use, as long as appropriate credit is given, a link to the license is provided, and any changes made to the material are indicated.

For this project, the dataset was downloaded and processed locally for analysis, modeling, and dashboard development. The raw dataset files are **not included** in the GitHub repository. Instead, the repository provides the project code, SQL scripts, notebooks, documentation, dashboard screenshots, and analytical methodology. Users who want to reproduce the project should download the dataset directly from the original source.

Dataset source:

FreshRetailNet-50K by Dingdong-Inc

<https://huggingface.co/datasets/Dingdong-Inc/FreshRetailNet-50K>

License:

Creative Commons Attribution 4.0 International (CC BY 4.0)

<https://creativecommons.org/licenses/by/4.0/>

License scope note:

The project code, notebooks, SQL scripts, documentation, and original analysis may be licensed separately by the project owner. However, the FreshRetailNet-50K dataset remains subject to its original **CC BY 4.0** license. The raw dataset files are not redistributed in this repository, and no ownership is claimed over the original dataset.

7. Data Understanding

7.1 Dataset Files

The FreshRetailNet-50K dataset was provided in Parquet format and processed locally for analysis, SQL preparation, and Power BI reporting. The dataset contains two main files:

- train.parquet
- eval.parquet

For practical project execution, the Parquet files were converted locally into CSV format to support easier integration with SQL-based validation workflows and Power BI dashboard development. The converted files were used only for local analysis and are not intended to be redistributed through the project repository.

The training dataset contains **4,500,000 rows and 19 columns**, while the evaluation dataset contains **350,000 rows and 19 columns**. The training period covers data from **2024-03-28 to 2024-06-25**, while the evaluation period covers **2024-06-26 to 2024-07-02**. This structure supports a time-based project workflow where the training data is used for historical analysis, feature engineering, model development, and backtesting, while the evaluation data can be used for forecast validation or future-period assessment.

7.2 Schema

The dataset contains a structured combination of store identifiers, product hierarchy fields, demand measures, stockout indicators, promotional variables, holiday/activity flags, and weather-related contextual variables.

The schema consists of the following 19 fields:

city_id

store_id

management_group_id

first_category_id

second_category_id

third_category_id

product_id

dt

sale_amount

hours_sale

stock_hour6_22_cnt

hours_stock_status

discount

holiday_flag

activity_flag

precpt

avg_temperature

avg_humidity

avg_wind_level

The fields can be grouped into several analytical categories:

Store and Location Fields

- city_id
- store_id

These fields identify the city and store where the product demand observation was recorded.

Product Hierarchy Fields

- management_group_id
- first_category_id
- second_category_id
- third_category_id
- product_id

These fields represent the product hierarchy, allowing analysis at diverse levels of product aggregation, from broad management groups down to individual product IDs.

Time Field

- dt

This field represents the date of the observation and is the main time dimension used for trend analysis, seasonality analysis, forecasting, and backtesting.

Demand Fields

- sale_amount
- hours_sale

The sales_amount field represents the daily sales amount after normalization. The hours_sale field contains hourly sales sequence information. After conversion from Parquet to CSV, this sequence field is stored as a string representation and would require parsing if hourly-level analysis is needed.

Stockout Fields

- stock_hour6_22_cnt
- hours_stock_status

The stock_hour6_22_cnt field represents the number of out-of-stock hours between 6:00 and 22:00. The hours_stock_status field contains hourly stockout status information. Similar to hours_sale, this field is stored as a string after CSV conversion and can be parsed later for hourly stockout analysis.

Business Context Fields

- discount
- holiday_flag
- activity_flag

These fields support analysis of discount behavior, holiday effects, and activity or promotional periods.

Weather Fields

- precpt
- avg_temperature
- avg_humidity
- avg_wind_level

These fields provide weather-related context that can be used to evaluate external demand drivers and improve forecasting features.

7.3 Grain

The dataset is structured at the **product-store-date level**. This means that each row represents one observation for a specific product in a specific store on a specific date.

The expected analytical grain is:

one row = one product × one store × one date

A grain validation check was performed using the combination of:

store_id + product_id + dt

The result showed **0 duplicate rows** at this level, confirming that the dataset is correctly structured for daily product-store demand analysis.

This grain is important because it defines how the main project metrics should be calculated. Demand forecasting, stockout risk assessment, product performance, store performance, and inventory risk segmentation should all be based on this product-store-date structure unless the data is intentionally aggregated to a higher level such as product, store, city, or category.

7.4 Key Columns

The most important columns for this project are directly aligned with the project objectives of demand forecasting, stockout risk analysis, promotion impact evaluation, and inventory intelligence.

city_id

Represents the encoded city identifier. The training dataset contains **18 unique cities**, which allows demand and stockout patterns to be compared geographically.

store_id

Represents the encoded store identifier. The training dataset contains **898 unique stores**, making it possible to analyze demand behavior and stockout exposure at store level.

product_id

Represents the encoded product identifier. The training dataset contains **865 unique products**, allowing product-level demand forecasting, product performance analysis, and inventory risk segmentation.

Product Hierarchy Columns

The dataset contains multiple product hierarchy fields:

management_group_id

first_category_id

second_category_id

third_category_id

The training dataset includes:

- **7 management groups**
- **32 first-level categories**
- **84 second-level categories**
- **233 third-level categories**

These fields allow the project to analyze demand not only at product level, but also across broader category structures.

dt

Represents the date of the observation. The training data covers the period from **2024-03-28 to 2024-06-25**, while the evaluation data covers **2024-06-26 to 2024-07-02**. This time structure supports time series analysis, feature engineering, and forecast evaluation.

sale_amount

Represents the normalized daily sales amount. This is the primary demand target used for trend analysis, forecasting, and product-store demand evaluation.

hours_sale

Contains hourly sales sequence information. This field can support deeper intraday demand analysis if parsed from its stored string representation after CSV conversion.

stock_hour6_22_cnt

Represents the number of out-of-stock hours between 6:00 and 22:00. This is a core field for stockout risk analysis and inventory availability monitoring.

hours_stock_status

Contains hourly stockout status information. This field is useful for more granular stockout analysis, especially if the project later expands into hourly stockout behavior or latent demand analysis.

discount

Represents the discount level applied to the product. This field supports analysis of discount impact, promotion sensitivity, and demand uplift behavior.

holiday_flag

Indicates whether the observation date is associated with a holiday. This field supports holiday demand analysis and calendar-based forecasting features.

activity_flag

Indicates whether the observation is associated with an activity or promotional event. This field can help evaluate the effect of business activity periods on demand.

Weather Columns

The weather-related fields include:

precpt

avg_temperature

avg_humidity

avg_wind_level

These variables provide external context that may help explain demand variation and improve forecasting model performance.

7.5 Analytical Relevance

The FreshRetailNet-50K dataset is highly relevant to the project because it combines demand, inventory availability, promotional context, calendar effects, and weather variables in a single product-store time series structure.

From a **Data Analyst** perspective, the dataset supports:

- demand trend analysis.
- seasonality analysis
- store-level performance analysis.
- product and category performance analysis
- discount and activity impact analysis.
- holiday demand comparison.
- stockout pattern analysis
- weather-related demand exploration
- Power BI dashboard development

From a **Data Scientist** perspective, the dataset supports:

- time series forecasting
- Lag feature creation
- rolling window feature engineering
- stockout-aware feature development
- discount and weather feature integration
- baseline model comparison

- machine learning forecasting models
- time-based backtesting and evaluation

The dataset is especially valuable because it includes stockout-related indicators. This is important because observed sales may not fully represent true customer demand when products are unavailable. By including stockout information, the project can move beyond simple sales forecasting and incorporate inventory risk analysis into the forecasting workflow.

The absence of missing values across all 19 fields also provides a strong starting point for analysis. No missing values were detected in the training dataset, meaning the initial data structure is complete and suitable for downstream analysis, feature engineering, and model development. However, further validation is still required in the Data Quality Assessment phase to check temporal continuity, value ranges, business logic, and consistency of stockout-related fields.

Overall, the dataset provides a solid foundation for building an end-to-end **Demand Forecasting, Stockout Risk, and Inventory Intelligence** solution. Its combination of product-store granularity, time-based structure, stockout indicators, discount variables, holiday/activity flags, and weather features make it suitable for both business intelligence reporting and predictive modeling.

8. Methodology

This project follows an end-to-end analytics and forecasting methodology designed to transform raw fresh retail time series data into business-ready insights, predictive forecasting outputs, and inventory risk indicators. The methodology combines SQL, Python, and Power BI to cover the full workflow from data understanding and preparation to exploratory analysis, feature engineering, forecasting, evaluation, dashboarding, and business recommendation development.

The workflow was structured around one main principle: demand in fresh retail should not be analyzed using sales history alone. Because observed sales can be affected by stockouts, discounts, holidays, promotional activity, and weather conditions, the project treats demand forecasting as both a predictive modeling problem and an inventory intelligence problem. This approach allows the analysis to distinguish between low sales caused by weak demand and low sales potentially caused by product unavailability.

The methodology was organized into the following stages:

1. Data Source Review and Local Preparation

The project began by reviewing the FreshRetailNet-50K dataset structure, source, license, and available files. The original data was provided in Parquet format through two main files: `train.parquet` and `eval.parquet`. For local project execution, the files were converted into CSV format to support easier inspection, SQL loading, and Power BI integration. The raw and converted dataset files are used locally only and are not redistributed through the GitHub repository.

2. Data Understanding

The first analytical step focused on understanding the dataset structure, including row counts, column names, data types, time coverage, unique stores, unique products, product hierarchy levels, and the analytical grain. The dataset was confirmed to be structured at the product-store-date level, where each row represents one product sold in one store on one date. This grain is central to the project because demand forecasting, stockout analysis, and inventory risk segmentation all depend on correct product-store-date interpretation.

3. SQL-Based Data Validation and Preparation

SQL is used to create a structured preparation and validation layer. This includes basic table loading, row count checks, date range validation, duplicate grain checks, product/store/category profiling, and summary queries. SQL also supports the creation of analytical views that can later be used in Power BI for reporting. The SQL layer helps ensure that the data is organized and validated before deeper analysis and dashboarding.

4. Data Quality Assessment

A data quality assessment is performed to evaluate whether the dataset is suitable for analysis and forecasting. The assessment covers missing values, duplicate records, date consistency, grain validation, value ranges, categorical flag validity, and stockout field consistency. Although the initial review showed no missing values across the training dataset, further checks are required to confirm temporal continuity, valid discount values, valid stockout hour counts, and consistency between daily and hourly stockout fields.

5. Exploratory Data Analysis

Exploratory Data Analysis is used to understand historical demand behavior and business patterns. The EDA stage examines demand trends over time, product and store performance, city-level demand variation, product category contribution, sales volatility, discount behavior, holiday effects, activity flag impact, weather-related demand variation, and stockout patterns. This stage provides the business foundation for later forecasting and inventory risk modeling.

6. Feature Engineering

Feature engineering transforms the product-store-date data into a forecasting-ready structure. The project creates features such as lag variables, rolling averages, rolling volatility measures, calendar features, stockout indicators, discount variables, activity and holiday flags, and weather inputs. These engineered features allow machine learning models to learn from historical demand patterns, recent demand momentum, product-store behavior, and external demand drivers.

7. Forecasting Model Development

The forecasting phase starts with simple baseline models to establish benchmark performance. These include naive forecasting, moving average forecasting, and seasonal naive forecasting. After evaluating baseline performance, the project progresses to machine learning models such as XGBoost or LightGBM using engineered features. The forecasting target is `sale\_amount`, which represents normalized daily demand.

8. Backtesting and Model Evaluation

Because this is a time series forecasting problem, model evaluation is performed using time-based validation rather than random train-test splitting. Backtesting is used to simulate how the model would perform when forecasting future periods based only on past data. Forecasting performance is evaluated using metrics such as MAE, RMSE, WAPE, and forecast bias. These metrics help assess both the size and direction of forecasting errors.

9. Inventory Risk Framework

The project connects forecasting outputs with inventory risk interpretation. Forecasted demand, recent sales behavior, stockout hours, demand volatility, discount sensitivity, and product-store trends are combined to assign products and stores into inventory risk segments. These segments may include high stockout risk, potential overstock risk, stable demand, promo-sensitive demand, and products requiring close replenishment monitoring.

10. Power BI Dashboard Development

The final analytical outputs are translated into an interactive Power BI dashboard. The dashboard is designed to support business users by presenting demand trends, product, and store performance, stockout exposure, discount and activity impact, weather-related demand patterns, forecast results, and model performance. The dashboard acts as the business-facing layer of the project and helps convert technical analysis into actionable decision support.

11. Insight Generation and Business Recommendations

The final stage of the methodology translates analytical findings into business recommendations. These recommendations focus on improving replenishment prioritization, monitoring high-risk products, reducing stockout exposure, identifying promo-sensitive products, and using forecasting outputs to support proactive inventory planning. The goal is to move from reactive sales reporting to forward-looking demand and inventory intelligence.

Overall, this methodology provides a complete workflow that connects data preparation, business analysis, forecasting, inventory risk assessment, and dashboard reporting. It demonstrates how SQL, Python, and Power BI can be used together to build an integrated analytics solution for fresh retail demand planning.

9. Data Integration and Preparation

9.1 Parquet to CSV Conversion and SQL Loading

The FreshRetailNet-50K dataset was originally provided in Parquet format through two main files: train.parquet and eval.parquet. To support the project workflow across SQL validation, analytical preparation, and Power BI reporting, the original Parquet files were converted locally into CSV format. This conversion was performed only for local project execution and data processing purposes; the raw dataset files are not redistributed as part of the project repository.

After converting the files, a daily analytical version of the dataset was prepared to support the main analysis layer. The original dataset includes both daily-level variables and hourly sequence fields, such as hourly sales and hourly stock status. Since the first stage of the project focuses on daily demand analysis, SQL validation, Power BI reporting, and forecasting preparation, the initial SQL-ready version retained the daily analytical fields and excluded the hourly sequence fields from this preparation layer.

The prepared daily training dataset contains **4,500,000 rows and 17 columns**, while the prepared daily evaluation dataset contains **350,000 rows and 17 columns**. The training dataset covers the period from **2024-03-28 to 2024-06-25**, while the evaluation dataset covers **2024-06-26 to 2024-07-02**. This creates a clear time-based structure where the training period can be used for historical analysis and model development, while the evaluation period can support future-period validation and forecasting assessment.

The prepared CSV files were then loaded into a MySQL database named freshretail_db. Two tables were created to store the daily analytical data:

- freshretail_daily_train
- freshretail_daily_eval

After loading the data into MySQL, SQL validation checks were performed to confirm that the import process preserved the expected structure and row counts. The validation confirmed that the training table contains **4,500,000 rows, 18 unique cities, 898 unique stores, and 865 unique products**. The product hierarchy was also preserved, including **7 management groups, 32 first-level categories, 84 second-level categories, and 233 third-level categories**.

Additional SQL validation confirmed that the training table covers the expected date range from **2024-03-28 to 2024-06-25**. The main numeric fields were also validated: sale_amount ranges from **0 to 44.9**, stock_hour6_22_cnt ranges from **0 to 16**, and discount ranges from **0 to 1.088**. These results confirmed that the SQL-loaded data matched the prepared analytical dataset and that no structural mismatch or data loss occurred during the loading process.

Overall, the Parquet-to-CSV conversion and SQL loading process successfully produced a validated daily analytical data layer. This layer serves as the foundation for the next stages of the project, including SQL-based preparation, data quality assessment, exploratory data analysis, Power BI dashboard development, feature engineering, and forecasting model development.

9.2 SQL Preparation Layer

After the prepared daily analytical files were loaded into MySQL, a SQL preparation layer was created to organize the data into reusable analytical views. The purpose of this layer was to move beyond raw table storage and build structured SQL outputs that could support validation, exploratory analysis, Power BI reporting, and later inventory risk analysis.

The SQL preparation layer was built on top of the validated daily training table, `freshretail_daily_train`, which contains **4,500,000 rows**, **18 cities**, **898 stores**, **865 products**, and a product hierarchy consisting of **7 management groups**, **32 first-level categories**, **84 second-level categories**, and **233 third-level categories**. The table covers the training period from **2024-03-28 to 2024-06-25**.

The first view created was `vw_daily_sales_base`, which acts as the main reusable base view for the prepared daily dataset. This view preserves the validated product-store-date grain and includes the core fields required for analysis, such as product hierarchy, store identifiers, date, sales amount, stockout hours, discount, holiday flag, activity flag, and weather variables. This base view provides a consistent foundation for all downstream SQL summaries.

Several analytical views were then created to support different business and reporting needs. The `vw_product_performance` view summarizes demand and stockout behavior at the product level. It includes metrics such as total observations, total sales, average daily sales, maximum daily sales, active sales days, total stockout hours, average stockout hours, and average discount. This view will support product performance analysis, ABC analysis, demand concentration review, and later product-level inventory risk segmentation.

The `vw_store_performance` view summarizes sales and stockout behavior at the store level. It provides store-level metrics such as total sales, average daily sales, active products, total observations, total stockout hours, average stockout hours, and average discount. This view will support store comparison, city-level operational analysis, and Power BI visuals focused on store performance and stock availability.

A `vw_daily_summary` view was created to aggregate demand and inventory indicators by date. This view supports time-series trend analysis by summarizing daily total sales, active stores, active products, total stockout hours, average stockout hours, and average discount. The daily summary confirmed that each training date contains **50,000 observations**, with **898 active stores** and **865 active products**, indicating a consistent daily product-store analytical structure.

To support executive-level reporting, a `vw_monthly_summary` view was also created. This view aggregates the daily data into monthly summaries, including total observations, active stores, active products, total sales, average sales per observation, total stockout hours, average stockout hours, and average discount. This monthly layer is useful for high-level Power BI reporting and management review, while the daily layer remains the primary structure for forecasting and operational analysis.

A stockout-focused view, `vw_product_store_stockout_summary`, was created at the product-store level. This view combines product, store, sales, and stockout indicators to identify where product availability issues are concentrated. It includes total days, total stockout hours, average stockout hours, stockout days, total sales, and average daily sales. This view is especially important for the inventory risk framework because it helps identify products and stores with repeated stockout exposure.

Finally, a `vw_discount_summary` view was created to prepare the dataset for discount impact analysis. This view groups observations by discount level and summarizes total observations, total sales, average sales, total stockout hours, and average stockout hours. This will support later analysis of whether discount levels are associated with higher demand, increased stockout exposure, or different sales behavior.

Overall, the SQL preparation layer created a structured analytical foundation for the project. It transformed the loaded daily dataset into reusable business-ready views that support product analysis, store analysis, daily and monthly trend reporting, stockout monitoring, discount impact analysis, and Power BI dashboard development. This layer also improves project reproducibility by ensuring that key analytical summaries can be regenerated consistently from the validated MySQL tables.

9.3 Python Preparation Layer

After completing the Parquet-to-CSV conversion, SQL loading, and SQL preparation views, a Python preparation layer was created to support the next stages of the project workflow. The purpose of this layer is to make the prepared daily analytical data ready for Python-based data quality assessment, exploratory data analysis, feature engineering, forecasting model development, and model evaluation.

The Python preparation layer uses the daily analytical files created during the ingestion process:

- `freshretail_daily_train.csv`
- `freshretail_daily_eval.csv`

These files represent the simplified daily product-store-date analytical structure prepared from the original FreshRetailNet-50K data. The training file contains **4,500,000 rows**, while the evaluation file contains **350,000 rows**. The training period covers **2024-03-28 to 2024-06-25**, and the evaluation period covers **2024-06-26 to 2024-07-02**. This structure provides a clear time-based split between historical training data and a future evaluation period.

During this preparation stage, the date column `dt` was converted into a proper datetime format. This step is essential because the project depends on time-based analysis, chronological sorting, trend evaluation, backtesting, and forecasting. Without correct date formatting, later stages such as lag feature creation, rolling window calculations, and time-based validation would not be reliable.

The data was then sorted by `store_id`, `product_id`, and `dt`. This sorting step ensures that each product-store time series is arranged in chronological order. This is especially important for forecasting workflows because future feature engineering will rely on historical values, such as previous-day sales, previous-week sales, rolling averages, and recent stockout behavior.

A grain validation check was also performed in Python to confirm that the prepared data remained structured at the expected analytical level:

one row = one store × one product × one date

The validation confirmed that there were **0 duplicate rows** at the `store_id + product_id + dt` grain in both the training and evaluation datasets. This confirms that the prepared Python layer maintains the same clean product-store-date structure validated earlier during SQL checks.

The final data type review confirmed that the dataset is suitable for downstream analysis. Identifier fields such as `city_id`, `store_id`, `product_id`, and product hierarchy fields are stored as integer values. The target variable `sale_amount` is stored as a numeric field, while `stock_hour6_22_cnt` is stored as an integer field representing daily stockout exposure. Business context variables such as `discount`, `holiday_flag`, and `activity_flag`, along with weather variables such as precipitation, temperature, humidity, and wind level, are also preserved for later analysis and modeling.

This Python preparation layer intentionally focuses on structural readiness rather than advanced feature creation. At this stage, no forecasting features such as lag variables, rolling averages, or target transformations were created. Those transformations will be managed later in the **Feature Engineering** phase. The goal here is to create a stable, clean, and chronologically ordered dataset that can serve as the foundation for those future modeling steps.

Prepared local versions of the dataset were saved after date conversion, sorting, and validation. These files will be used in later stages of the project, including data quality assessment, EDA, feature engineering, forecasting, and model evaluation. Because these files are derived from the original dataset, they are intended for local project execution only and should not be redistributed through the GitHub repository.

Overall, the Python preparation layer completes the data ingestion and preparation workflow by creating a clean, structured, and time-aware analytical dataset. Together with the SQL validation and SQL preparation views, this layer ensures that the project can move confidently into deeper data quality checks, exploratory analysis, and forecasting development.

10.7 Train vs Eval Consistency

The prepared training and evaluation datasets were assessed to verify structural, temporal, and entity-level consistency before proceeding to exploratory analysis, feature engineering, and forecasting model development.

The prepared training dataset contains **4,500,000 observations and 17 columns**, while the prepared evaluation dataset contains **350,000 observations and 17 columns**. A schema comparison confirmed that both datasets share the exact same set of analytical fields and column order, ensuring full compatibility between the training and evaluation layers.

A data type consistency check showed that all variables have matching data types across both datasets. Identifier fields are stored as integers, numerical measures such as sales, stockout hours, discount values, and weather variables are stored as numeric fields, and the `dt` field was converted to a datetime format in both datasets. No data type mismatches were detected.

The temporal structure of the datasets was also validated. The training dataset covers the period from **2024-03-28 to 2024-06-25**, while the evaluation dataset covers **2024-06-26 to 2024-07-02**. The evaluation period begins immediately after the training period ends, creating a one-day transition with no overlap between the two datasets. This chronological separation prevents data leakage and supports reliable future-period forecasting evaluation.

Entity coverage was fully consistent across both datasets. The training and evaluation datasets each contain:

- **18 unique cities**
- **898 unique stores**
- **865 unique products**

No new cities, stores, or products were introduced in the evaluation period. This confirms that all evaluation entities were already observed during training and therefore possess historical information that can be used for future modeling and forecasting.

A product-store coverage assessment was performed to compare the analytical universe represented in both datasets. The results showed that both train and evaluation datasets contain exactly **50,000 unique product-store combinations**. No new product-store pairs were found in the evaluation period, and no training product-store combinations were missing from the evaluation dataset. This indicates complete alignment between the two datasets at the analytical grain used throughout the project.

The evaluation dataset was also assessed for time-series continuity. The evaluation period spans **7 consecutive days**, and every product-store series contains the expected seven daily observations. As a result, all **50,000 product-store series were complete**, with **0 incomplete evaluation series** detected.

Overall, the consistency assessment confirmed that the prepared training and evaluation datasets are fully aligned in terms of schema, data types, temporal structure, entity coverage, product-store coverage, and time-series completeness. These results provide a reliable and well-structured foundation for the subsequent stages of the project, including exploratory data analysis, feature engineering, forecasting model development, and forecast evaluation.

10.8 Data Quality Summary

A comprehensive data quality assessment was conducted on the prepared daily analytical datasets to evaluate their suitability for exploratory analysis, feature engineering, forecasting model development, inventory risk assessment, and Power BI reporting.

The assessment confirmed that the prepared datasets are structurally reliable and ready for downstream analysis. Both the training and evaluation datasets contain complete records across all analytical fields, with **no missing values detected**. This indicates that the data preparation process successfully produced fully populated datasets suitable for analysis and modeling.

Duplicate validation was performed at the main analytical grain of the project:

Store × Product × Date

The results confirmed that there are **no duplicate records** in either the training or evaluation dataset at this grain. Therefore, each row represents one unique product-store-date observation. This is important because duplicate records could inflate sales, stockout hours, and other analytical metrics, leading to misleading insights or inaccurate forecasting results.

The date continuity assessment showed that the training dataset covers a complete **90-day period** from **2024-03-28 to 2024-06-25**. Each of the **50,000 product-store time series** contains exactly **90 daily observations**, resulting in **100% complete time series coverage**. No incomplete product-store series were detected in the training dataset.

The evaluation dataset also demonstrated full time-series continuity. It covers a continuous **7-day future period** from **2024-06-26 to 2024-07-02**, with all **50,000 product-store combinations** containing the expected 7 daily observations. This makes the evaluation dataset suitable for future-period validation and forecasting assessment.

Numeric value validation confirmed that the main analytical fields contain reasonable and logically consistent values. The `sale_amount` field ranges from **0 to 44.9**, with **no negative sales values** detected. This confirms that the sales target variable is valid for analysis and forecasting.

The `stock_hour6_22_cnt` field ranges from **0 to 16**, which is consistent with the expected stockout monitoring window between **06:00 and 22:00**. No invalid stockout hour values were found. This confirms that the stockout field is logically consistent and suitable for inventory risk analysis.

The discount field ranges from **0 to 1.088**, with no negative discount values detected. Most observations fall within the expected range, while only **34 observations** have discount values above 1. Since this represents a very small share of the overall dataset, these records were not treated as a major data quality issue, but they should be monitored during exploratory analysis.

Weather-related variables also showed reasonable ranges. The precipitation, average temperature, humidity, and wind-level fields did not show obvious invalid values during the range validation step. These variables were therefore retained for later analysis of external demand drivers and potential forecasting features.

Binary flag validation confirmed that both `holiday_flag` and `activity_flag` contain only valid binary values: **0 and 1**. No invalid flag values were detected. In the training dataset, approximately **34.44%** of observations were associated with holidays, while approximately **37.84%** were associated with activity periods. These fields are suitable for later analysis of holiday effects, activity impact, and demand variation.

A dedicated analysis was also conducted to examine the relationship between zero sales and stockout behavior. The training dataset contains **200,760 zero-sales observations**, representing **4.46%** of the full training dataset. Among these zero-sales records, **153,816 observations**, or **76.62%**, also had stockout hours.

This finding is important because it shows that many zero-sales cases may not represent weak customer demand. Instead, a large share of zero-sales records may be related to product unavailability. Zero-sales

observations had an average of **12.03 stockout hours**, compared with only **2.77 stockout hours** for observations with positive sales. This suggests that stockouts may be a major contributor to zero-sales behavior.

This insight is especially important for the forecasting phase. Treating all zero-sales observations as true zero demand could lead to demand underestimation, especially for products that were unavailable for a sizable portion of the day. Therefore, stockout indicators should be considered during feature engineering, forecasting, and inventory risk segmentation.

The prepared training and evaluation datasets were also checked for consistency. Both datasets contain the same **17 analytical columns**, matching data types, and the same business entity universe. Both train and eval contain **18 cities**, **898 stores**, **865 products**, and **50,000 product-store combinations**. No new cities, stores, products, or product-store pairs appear in the evaluation dataset.

The evaluation period starts immediately after the training period, with no overlap between train and eval dates. This confirms that the evaluation dataset can be used as a clean future-period dataset for validation and forecasting assessment without data leakage.

Overall, the data quality assessment confirms that the prepared datasets are complete, non-duplicated, temporally continuous, logically consistent, and suitable for the next stages of the project. The data provides a solid foundation for exploratory data analysis, feature engineering, forecasting model development, inventory risk analysis, and business dashboard reporting.

11. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand the historical demand behavior, product-store performance, stockout exposure, and key business drivers within the prepared FreshRetailNet-50K daily dataset. The purpose of this stage is to move beyond data validation and begin identifying meaningful business patterns that can support inventory planning, demand forecasting, and operational decision-making.

The EDA phase focuses on analyzing demand from multiple perspectives, including overall daily sales trends, seasonality, product performance, store performance, category-level contribution, stockout behavior, discount impact, holiday and activity effects, and weather-related demand variation. These analyses help establish a clear understanding of how demand changes over time and which factors may influence sales performance across products and stores.

A key objective of this section is to identify patterns that are relevant for both business reporting and forecasting model development. From a business perspective, the analysis helps highlight high-demand periods, products with strong or weak sales performance, stores with higher stockout exposure, and categories that contribute most to total demand. From a modeling perspective, the EDA helps determine which variables may be useful for feature engineering, such as lag features, rolling averages, discount indicators, holiday flags, activity flags, weather variables, and stockout-related features.

Stockout behavior is treated as a central part of the analysis because observed sales may not always represent true customer demand. When products are unavailable, recorded sales can be artificially low, leading to potential demand underestimation. Therefore, the EDA investigates not only sales patterns, but also how stockout exposure interacts with zero sales, demand variation, and product-store performance.

The insights generated from this EDA stage will guide the later stages of the project, including feature engineering, forecasting model selection, backtesting, inventory risk segmentation, and Power BI dashboard design. Overall, the EDA provides the analytical foundation needed to connect historical demand patterns with forward-looking forecasting and inventory intelligence.

11.1 Overall Demand Trends

The overall demand trend analysis was conducted to understand how total sales, average sales, and stockout exposure changed over the training period.

The prepared training dataset contains a consistent daily structure, with **50,000 observations per day**, covering **865 products** across **898 stores**. This consistency confirms that the daily analytical layer is well-structured for time series analysis and future forecasting work.

Across the full training period, total sales reached approximately **4.49 million normalized sales units**. The average daily total sales were approximately **49,929.56**, while daily total sales ranged from a minimum of approximately **34,893.48** to a maximum of approximately **72,938.82**. The average sales per product-store observation were approximately **0.999**.

The daily trend suggests noticeable demand variation across the training period. Sales were relatively lower at the beginning of the period, with daily totals around **35K** in late March. Toward the later part of the training period, demand increased, with several days in June exceeding **70K** in total daily sales. This indicates that the dataset contains meaningful temporal demand variation that should be considered during feature engineering and forecasting.

Stockout exposure was also significant. The dataset recorded approximately **14.31 million total stockout hours** across the training period, with an average of approximately **159,017 stockout hours per day**. This confirms that stockout behavior is a major operational factor in the dataset and should not be ignored during demand interpretation, inventory risk analysis, or forecasting model development.

Overall, the demand trend analysis shows that the dataset contains stable daily coverage, meaningful variation in sales, and substantial stockout exposure. These findings support the need for a forecasting and inventory intelligence framework that considers both historical demand patterns and product availability constraints.

11.2 Seasonality Analysis

Seasonality analysis was conducted to understand how demand and stockout behavior vary across different calendar and business time dimensions. The analysis focused on day-of-week patterns, weekday versus weekend behavior, holiday effects, and activity-period demand differences.

The day-of-week analysis showed that demand is not evenly distributed across the week. Weekend days recorded the highest average sales per observation, with **Sunday** showing the highest average sales at approximately **1.20**, followed by **Saturday** at approximately **1.18**. In contrast, lower average sales were observed during mid-week days, particularly **Thursday** and **Wednesday**.

The weekday versus weekend comparison confirmed this pattern more clearly. Weekday observations recorded an average sales value of approximately **0.92**, while weekend observations recorded a higher average sales value of approximately **1.19**. This indicates that demand is stronger during weekends. Weekend observations also showed higher average stockout hours, suggesting that higher weekend demand may create additional pressure on product availability.

Holiday analysis also showed a clear demand difference. Non-holiday observations recorded an average sales value of approximately **0.91**, while holiday observations recorded a higher average sales value of approximately **1.16**. This suggests that holidays are associated with stronger demand. Holiday observations also had slightly higher average stockout hours, indicating that holiday demand may increase the risk of product availability issues.

Activity-period analysis showed a smaller difference in average sales. Observations without activity periods recorded average sales of approximately **0.99**, while activity-period observations recorded average sales of approximately **1.01**. Although the sales uplift is relatively modest, activity-period observations had a much lower average discount factor, indicating that these periods are likely associated with stronger promotional or discount activity.

Overall, the seasonality analysis indicates that calendar-based patterns are important in this dataset. Weekends and holidays are associated with noticeably higher demand and higher stockout exposure. These findings suggest that calendar variables such as day of week, weekend indicator, holiday flag, and activity flag should be included in the feature engineering phase and considered during forecasting model development.

11.3 Product Performance Analysis

Product performance analysis was conducted to understand how demand, sales contribution, and stockout exposure vary across products in the prepared training dataset. This analysis is important because inventory and forecasting decisions are often made at the product level, especially in fresh retail environments where product availability, demand variability, and replenishment timing directly affect business performance.

The analysis showed that product demand is not evenly distributed across all products. A relatively small group of products contributes a large share of total sales, while many other products represent a long tail of lower-demand items. This pattern is important because it indicates that products should not all be managed with the same level of attention or forecasting priority.

The highest-selling products during the training period included product_id = 267, product_id = 4, product_id = 300, product_id = 117, and product_id = 215. Product 267 recorded the highest total sales at approximately **183,395**, followed closely by product 4 with approximately **183,265** total sales. Other strong contributors included product 300 with approximately **154,506**, product 117 with approximately **150,033**, and product 215 with approximately **144,031** total sales.

These products represent key demand drivers in the dataset and should receive higher attention in later stages of the project, especially during feature engineering, forecasting model development, replenishment prioritization, and Power BI dashboard design.

The stockout exposure analysis showed that stockout risk is also concentrated among specific products. Product 834 recorded the highest total stockout exposure, with approximately **467,586 stockout hours**. Other products with high stockout exposure included product 129 with approximately **292,575 stockout**

hours, product 580 with approximately **268,214**, product 600 with approximately **255,505**, and product 4 with approximately **212,618 stockout hours**.

This finding is highly relevant to the inventory intelligence objective of the project. Products with high stockout hours may represent availability risks, especially when they also generate high sales. These products may require closer monitoring, better replenishment planning, and potentially higher forecasting priority.

Several products appeared in both the top-sales and top-stockout rankings. Examples include products 4, 580, 834, 215, 596, 740, 300, 117, and 191. These products are particularly important because they combine meaningful demand contribution with high stockout exposure. In business terms, these products may represent situations where customer demand exists, but product availability may be limiting observed sales.

The product sales concentration analysis further confirmed that demand is concentrated across a relatively small portion of the product portfolio. The cumulative sales contribution curve rises steeply for the top-ranked products and then gradually flattens across the remaining products. This indicates that the highest-selling products account for a disproportionately large share of total sales.

To translate this concentration pattern into a business-ready segmentation, ABC classification was applied based on cumulative sales contribution. The results classified products into three groups:

- **A products:** 121 products contributing approximately **79.86%** of total sales
- **B products:** 167 products contributing approximately **15.10%** of total sales
- **C products:** 577 products contributing approximately **5.04%** of total sales

This means that only **121 out of 865 products** account for nearly **80% of total sales**, while the majority of products contribute a much smaller share of total demand. This is an important insight for inventory and forecasting strategy.

A-class products should receive the highest priority in forecasting accuracy, replenishment monitoring, and stockout risk management. These products have the greatest business impact and require tighter operational control. B-class products represent a moderate-priority group that should be monitored regularly but may not require the same level of attention as A products. C-class products represent the long tail of lower-contribution products and may be managed with simpler inventory policies unless specific products show high stockout exposure or strategic importance.

Overall, the product performance analysis confirms that product-level prioritization is essential for this project. The combination of sales contribution, stockout exposure, and ABC classification provide a solid foundation for later stages of the project, including product-level forecasting, inventory risk segmentation, replenishment prioritization, and Power BI dashboard development.

Product Sales vs Stockout Relationship

To better understand whether stockout exposure is associated with low sales or high demand pressure, a product-level relationship analysis was conducted between total sales and total stockout hours.

The analysis showed a strong positive relationship between product sales and stockout exposure. The correlation between total_sales and total_stockout_hours was approximately 0.72, indicating that higher-selling products generally tend to experience higher stockout hours. This suggests that stockout risk is not only a problem for low-performing products. In many cases, products with strong demand are also the products most exposed to availability pressure.

A scatter plot of total sales versus total stockout hours supported this finding. The chart showed a generally upward relationship, where products with higher stockout hours often also had higher total sales. However, the chart also revealed product-level variation, including products with high stockout exposure but relatively lower observed sales.

To make this analysis more actionable, products were segmented into four sales-stockout groups:

High Sales / High Stockout

High Sales / Low Stockout

Low Sales / High Stockout

Low Sales / Low Stockout

The most important segment was High Sales / High Stockout, which contained 408 products. This group contributed approximately 4.39 million total sales and accumulated approximately 13.89 million stockout hours. These products are highly important because they combine strong demand contribution with high product availability risk. They should receive priority attention in forecasting, replenishment planning, and inventory monitoring.

The Low Sales / High Stockout segment contained 25 products. Although these products contributed relatively low observed sales, they accumulated approximately 84,671 stockout hours. This group requires further investigation because low observed sales may be partially caused by product unavailability rather than weak customer demand. These products may represent cases of constrained or censored demand.

The results confirm that product performance should not be evaluated using sales alone. Stockout exposure provides critical context for interpreting observed demand. Products with high sales and high stockout exposure may require tighter replenishment control, while products with low sales and high stockout exposure may need further investigation to determine whether demand is being suppressed by availability constraints.

Overall, this analysis strengthens the inventory intelligence component of the project by linking demand contribution with product availability risk. It also provides a solid foundation for later inventory risk segmentation, where products can be prioritized based on both forecasted demand and recent stockout behavior.

11.4 Store Performance Analysis

Store performance analysis was conducted to understand how demand, sales contribution, and stockout exposure vary across store locations. This analysis is important because retail demand is not only product-driven, but also strongly affected by store location, customer behavior, and local operational conditions.

The analysis showed meaningful variation in performance across stores. The highest-sales stores included store 235, store 263, store 182, store 18, and store 555. Store 235 recorded the highest total sales at approximately **17,510**, followed by store 263 with approximately **17,408** and store 182 with approximately **17,270**. These stores represent important demand centers and should receive higher attention in store-level forecasting, replenishment planning, and dashboard monitoring.

Stockout exposure was also concentrated among specific stores. Store 18 recorded the highest total stockout hours at approximately **48,841**, followed by store 409 with approximately **46,671**, store 263 with approximately **46,330**, and store 182 with approximately **46,063**. Several of the highest-sales stores also appeared among the highest-stockout stores, indicating that stores with stronger demand may also face higher product availability pressure.

The relationship between store-level total sales and total stockout hours was strong. The correlation between `total_sales` and `total_stockout_hours` was approximately **0.96**, showing that high-sales stores tend to accumulate higher stockout exposure. This suggests that stockout risk is closely linked to store demand volume and operational scale.

However, average stockout hours showed a different pattern. While high-sales stores accumulated more total stockout hours, some lower-sales stores still showed relatively high average stockout pressure. This means store-level inventory risk should be evaluated using both total stockout hours and average stockout hours, rather than relying on total values only.

To make the store analysis more actionable, stores were segmented into four sales-stockout groups:

- **High Sales / High Stockout**
- **High Sales / Low Stockout**
- **Low Sales / High Stockout**
- **Low Sales / Low Stockout**

The most important segment was **High Sales / High Stockout**, which contained **406 stores**. This group generated approximately **3.11 million** total sales and approximately **9.57 million** stockout hours. These stores are commercially important and operationally risky, making them strong candidates for closer replenishment monitoring and inventory risk management.

The **Low Sales / High Stockout** segment contained **43 stores**, generating approximately **168,871** total sales and approximately **681,956** stockout hours. These stores require further investigation because low observed sales may be partially caused by product unavailability rather than weak customer demand alone.

City-level analysis showed that city_id = 0 dominated both total sales and total stockout exposure. This city generated approximately **2.37 million** total sales and approximately **7.14 million** stockout hours. However, this city also contains the largest number of stores, so city-level interpretation should consider both total values and average store-level performance metrics.

Overall, the store performance analysis confirms that store-level demand and stockout exposure are closely connected. High-sales stores often face higher stockout pressure and should receive stronger operational monitoring. At the same time, lower-sales stores with high stockout exposure should be reviewed to determine whether observed demand is being constrained by product availability issues.

11.5 Category Performance Analysis

Category performance analysis was conducted to understand how demand and stockout exposure vary across the product hierarchy. This analysis is important because business teams often manage assortment, replenishment, and performance monitoring not only at the individual product level, but also at broader category levels.

The dataset includes multiple product hierarchy levels: management_group_id, first_category_id, second_category_id, and third_category_id. Each level provides a different view of product performance, from broad product groups to more granular category structures. The analysis focused on identifying which category groups contribute most to total sales, which categories experience the highest stockout exposure, and whether high-demand categories are also associated with higher product availability risk.

At the management group level, management_group_id = 6 was the strongest contributor to total sales. This group included **367 products** and generated approximately **2.72 million** total sales. It also recorded approximately **7.46 million** stockout hours, the highest among all management groups. This indicates that management group 6 is both a major demand driver and a major source of stockout exposure.

Other important management groups included management_group_id = 5, which generated approximately **661K** total sales, and management_group_id = 2, which generated approximately **636K** total sales. However, the dominance of management group 6 shows that product hierarchy performance is not evenly distributed across the portfolio.

At the first-category level, first_category_id = 4 was the highest-performing category by total sales, generating approximately **1.16 million** total sales. This category also recorded approximately **3.14 million** stockout hours, making it the leading first-level category in both demand contribution and stockout exposure. This suggests that category 4 should receive high priority in forecasting, replenishment planning, and inventory monitoring.

Other high-performing first-level categories included first_category_id = 20, 16, 29, and 21. Category 20 generated approximately **639K** total sales, while category 16 generated approximately **472K** total sales. Category 29 generated approximately **462K** total sales and showed a particularly high stockout observation rate of approximately **65.13%**, indicating substantial availability pressure.

The second-category level provided a more detailed mid-level view of category behavior. This level sits between broad first-level categories and granular third-level categories, making it useful for identifying

category groups that are specific enough for operational monitoring but still broad enough for category-level planning.

At the second-category level, category path 4-28 was the strongest performer by total sales. It included **77 products** and generated approximately **997K** total sales. This same category also recorded the highest total stockout exposure, with approximately **2.30 million** stockout hours. This makes 4-28 a highly important category group from both a commercial and inventory risk perspective.

Other high-performing second-level category paths included 21-64, 20-50, 16-25, 29-76, and 29-78. Category 21-64 generated approximately **417K** total sales and had a high average sales value of approximately **2.32**, indicating strong demand intensity. Category 20-50 generated approximately **355K** total sales and approximately **901K** stockout hours, making it another important category group for stockout monitoring.

The second-level stockout analysis showed that some categories with moderate sales still carried high inventory risk. Category 16-25 recorded approximately **1.29 million** stockout hours, despite generating approximately **289K** total sales. Category 29-76 generated approximately **241K** total sales but recorded approximately **829K** stockout hours and a stockout observation rate of approximately **68.95%**. This suggests that some mid-level categories may face significant product availability pressure even when their total sales are not the highest.

The relationship between total sales and stockout exposure was especially strong at the second-category level. The correlation between `total_sales` and `total_stockout_hours` was approximately **0.94**, indicating that higher-selling second-level categories generally also experience higher stockout exposure. This confirms that inventory risk is strongly connected to demand volume at the mid-category level.

At the third-category level, the analysis provided a more granular view of demand concentration. The strongest third-level category by total sales was category path 4-28-81, which generated approximately **251K** total sales. Other major third-level category paths included 21-64-184, 29-76-60, 29-78-82, and 4-28-1. These granular categories represent important demand groups and should be considered in category-level reporting and forecasting prioritization.

Stockout exposure was also concentrated at the third-category level. Category path 29-76-60 recorded the highest total stockout exposure, with approximately **703K** stockout hours. This was followed by 28-72-154 with approximately **547K** stockout hours, and 4-28-81 with approximately **499K** stockout hours. These categories may represent important inventory risk areas because they combine substantial product activity with high stockout exposure.

The relationship between third-category sales and stockout exposure was also analyzed. The correlation between `total_sales` and `total_stockout_hours` at the third-category level was approximately **0.80**, indicating a strong positive relationship. This means that higher-selling granular categories generally tend to accumulate higher stockout hours. Therefore, stockout risk is not isolated to weak categories; it is strongly present in high-demand category groups.

To make the category analysis more actionable, third-level categories were segmented into four sales-stockout groups:

- **High Sales / High Stockout**
- **High Sales / Low Stockout**
- **Low Sales / High Stockout**
- **Low Sales / Low Stockout**

The most important segment was **High Sales / High Stockout**, which contained **113 categories** and **626 products**. This segment contributed approximately **4.38 million** total sales and approximately **13.74 million** stockout hours. These categories are commercially important and operationally risky, making them strong candidates for closer replenishment monitoring, forecasting attention, and inventory risk management.

The **Low Sales / High Stockout** segment contained only **4 categories** and **16 products**, generating approximately **11.7K** total sales and approximately **78.8K** stockout hours. Although this segment is small, it may include categories where observed sales are constrained by product availability. These categories should be reviewed further before being classified as low-demand categories.

Overall, the category performance analysis confirms that product hierarchy is highly relevant to both demand concentration and stockout risk. A relatively small number of category groups account for a large share of sales and stockout exposure. The analysis also shows that stock out risk increases with demand at multiple category levels, especially at the second category and third-category levels. These findings will help guide later stages of the project, including feature engineering, forecasting prioritization, inventory risk segmentation, Power BI dashboard development, and business recommendations.

11.6 Stockout Analysis

Stockout analysis was conducted to understand the scale, severity, and business impact of product availability issues across the prepared training dataset. This analysis is central to the project because observed sales may not always represent true customer demand. When a product is unavailable, recorded sales can be artificially low, creating a censored demand problem that must be considered during forecasting and inventory risk assessment.

The overall stockout summary showed that stockout exposure is a significant operational factor in the dataset. Out of **4.5 million product-store-date observations**, approximately **1.99 million observations** recorded at least one stockout hour. This represents approximately **44.27%** of all observations. Across the training period, total stockout exposure reached approximately **14.31 million stockout hours**, with an average of approximately **3.18 stockout hours** per observation.

The stockout field, `stock_hour6_22_cnt`, ranges from **0 to 16**, where 16 represents product unavailability across the full monitored period from 06:00 to 22:00. The median value was **0**, indicating that more than half of the observations did not experience stockout. However, the large total number of stockout observations confirms that availability risk is still widespread across the dataset.

To better understand severity, stockout observations were grouped into five buckets:

- **No Stockout:** 0 hours
- **Low Stockout:** 1–4 hours
- **Medium Stockout:** 5–8 hours
- **High Stockout:** 9–15 hours
- **Full-Day Stockout:** 16 hours

The severity distribution showed that **55.73%** of observations had no stockout. Low stockout cases represented approximately **13.66%** of observations, medium stockout cases represented approximately **15.55%**, high stockout cases represented approximately **11.01%**, and full day stockout cases represented approximately **4.04%**.

The relationship between stockout severity and sales showed an important pattern. Observations with **full-day stockout** had extremely low average sales of approximately **0.046**, and the zero-sales rate reached approximately **81.44%**. This indicates that when products are unavailable for the full monitored window, observed sales are strongly suppressed.

Low and medium stockout observations showed higher average sales than no-stockout observations. Low stockout observations recorded average sales of approximately **1.39**, while medium stockout observations recorded average sales of approximately **1.15**. This suggests that partial stockouts can occur in active demand situations, where products sell but become unavailable for part of the day. In contrast, full day stockouts are much more likely to prevent sales entirely.

The daily stockout trend also showed meaningful variation across the training period. Some days recorded total stockout hours above **200K**, while later periods showed lower daily stockout pressure around **118K–156K** on several days. This indicates that stockout exposure is not static over time and may be influenced by demand patterns, replenishment cycles, holidays, activity periods, or operational conditions.

At the product-store level, the analysis identified combinations with severe and repeated stockout exposure. The highest-risk product-store combination recorded **1,397 stockout hours** over 90 days, averaging approximately **15.52 stockout hours per day**. This same combination experienced stockout on **98.89%** of days. Several other top product-store combinations had stockout day rates above **90%**, indicating persistent availability problems.

These product-store combinations are especially important because inventory decisions are often made at the product-store level. A product may experience high stockout risk in one store but not in another. Therefore, product-store stockout analysis provides a more operationally actionable view than product-level or store-level analysis alone.

The row-level correlation analysis showed a slight negative relationship between `sale_amount` and `stock_hour6_22_cnt`, with a correlation of approximately **-0.10**. This suggests that, at the individual observation level, higher stockout hours are generally associated with lower observed sales. However, earlier product, store, and category analyses showed that high-demand products, stores, and categories often accumulate high total stockout exposure. This means stockout can be both a result of strong demand pressure and a cause of suppressed observed sales.

This distinction is important for the forecasting phase. Treating all low or zero sales as weak demand would be misleading, especially when those observations also include high stockout hours. Stockout indicators should therefore be incorporated into feature engineering and forecasting model development to help distinguish between true low demand and constrained demand caused by product unavailability.

Overall, the stockout analysis confirms that availability risk is a major factor in the dataset. Full-day stockouts strongly suppress observed sales, partial stockouts often occur during active demand situations, and severe product-store stockout patterns highlight operational areas that may require closer monitoring. These findings provide a solid foundation for the later Inventory Risk Framework, where demand strength, stockout exposure, and forecasted demand will be combined into actionable risk segments.

11.7 Discount, Activity, and Holiday Impact

Discount, activity, and holiday impact analysis was conducted to understand how business context variables are associated with demand behavior and stockout exposure. This analysis is important because promotional activity, price changes, and calendar events can influence customer demand and may create additional pressure on inventory availability.

The discount analysis showed a clear relationship between discount levels and observed sales. The dataset was grouped into discount buckets to make the results easier to interpret. The largest bucket was **Standard Price**, containing approximately **2.18 million observations**. This bucket recorded average sales of approximately **0.83**. In contrast, the **High Discount** bucket recorded the highest average sales at approximately **2.07**, suggesting that stronger discounting is associated with higher observed demand.

The relationship between discount and sales was also supported by the correlation analysis. The correlation between `sale_amount` and `discount` was approximately **-0.33**. Since lower discount values indicate stronger discounting, this negative correlation suggests that deeper discounts are associated with higher sales.

Discounting was also associated with stockout exposure. The **High Discount** bucket recorded an average of approximately **3.35 stockout hours** and a stockout observation rate of approximately **46.89%**. This indicates that stronger discounts may increase product movement and create greater product availability pressure. The **Medium Discount** bucket also showed meaningful stockout exposure, with an average of approximately **3.27 stockout hours** and a stockout observation rate of approximately **46.15%**.

The **Standard Price** bucket had the lowest average sales among the main discount groups, at approximately **0.83**, and had a zero-sales rate of approximately **6.05%**. This suggests that observations without stronger discounting may include a larger share of lower-demand or slower-moving product-store cases.

Activity-period analysis showed a more modest difference in average sales. Observations without activity periods recorded average sales of approximately **0.99**, while activity-period observations recorded average sales of approximately **1.01**. Although the direct sales difference was small, activity periods were strongly associated with lower average discount values. The average discount during activity periods

was approximately **0.80**, compared with approximately **0.98** during non-activity periods. This suggests that activity periods are closely linked to promotional pricing or stronger discounting.

Activity periods also showed slightly higher stockout exposure. Activity-period observations recorded average stockout hours of approximately **3.29**, compared with approximately **3.11** for non-activity observations. The stockout observation rate was also higher during activity periods, at approximately **46.50%**, compared with approximately **42.91%** during non-activity periods. This suggests that activity periods may put additional pressure on inventory availability, even if the direct uplift in average sales is modest.

Holiday analysis showed a clearer demand uplift. Holiday observations recorded average sales of approximately **1.16**, compared with approximately **0.91** for non-holiday observations. This indicates that holidays are associated with stronger demand in the dataset. Holiday observations also recorded slightly higher average stockout hours at approximately **3.31**, compared with approximately **3.11** for non-holiday observations. This suggests that holiday demand may contribute to higher product availability risk.

The combined holiday and activity analysis provided additional context. The highest average sales were observed in the **Holiday / Activity** segment, with average sales of approximately **1.16**. The **Holiday / No Activity** segment showed a very similar average sales value, also approximately **1.16**. In contrast, the non-holiday segments showed lower average sales, with **Non-Holiday / No Activity** at approximately **0.90** and **Non-Holiday / Activity** at approximately **0.93**. This indicates that holiday status appears to have a stronger demand effect than activity status alone.

The combined analysis also showed that **Holiday / Activity** observations had the highest average stockout hours at approximately **3.48** and the highest stockout observation rate at approximately **47.31%**. This suggests that when holidays and activity periods overlap, the business may face stronger product availability pressure.

Overall, the analysis indicates that discounting and holidays are meaningful demand drivers. Stronger discounts are associated with higher average sales and higher stockout exposure, while holidays are associated with clear demand uplift and slightly higher stockout pressure. Activity periods appear to be strongly linked to discounting, but their direct sales effect is smaller than the holiday effect. These findings support the inclusion of discount, holiday_flag, activity_flag, and interaction-based calendar features in the feature engineering and forecasting stages.

11.8 Weather Impact Analysis

Weather impact analysis was conducted to determine whether external weather conditions are associated with changes in sales demand, stockout exposure, or zero-sales behavior. The reviewed weather variables were precpt, avg_temperature, avg_humidity, and avg_wind_level.

The weather summary showed that precipitation ranged from **0** to **42.50**, with an average value of approximately **3.70**. Average temperature ranged from **12.06** to **30.88**, with a mean of approximately **22.28**. Average humidity ranged from **27.35** to **100**, with a mean of approximately **74.45**. Average wind level ranged from **0.97** to **3.85**, with a mean of approximately **1.72**. These ranges were considered reasonable and suitable for exploratory analysis and later feature engineering.

The correlation analysis showed that weather variables have relatively weak direct relationships with observed sales. The strongest positive relationship with `sale_amount` was observed for `avg_temperature`, with a correlation of approximately **0.056**. This was followed by `avg_humidity` at approximately **0.043** and `precpt` at approximately **0.037**. Average wind level showed almost no relationship with sales, with a correlation of approximately **-0.008**.

The relationship between weather variables and stockout exposure was also weak. Average temperature had a small negative correlation with `stock_hour6_22_cnt` of approximately **-0.063**, while precipitation, humidity, and wind level had near-zero relationships with stockout hours. This suggests that weather conditions may provide useful contextual signals, but they are not dominant standalone drivers of stockout behavior in this dataset.

To make the analysis more interpretable, weather variables were grouped into buckets. Temperature was divided into Low, Medium, and High Temperature segments. The results showed that average sales increased as temperature increased. Low Temperature observations recorded average sales of approximately **0.86**, Medium Temperature observations recorded average sales of approximately **0.99**, and High Temperature observations recorded average sales of approximately **1.07**. This indicates that warmer conditions may be associated with slightly higher demand.

Temperature also showed a difference in stockout exposure. Low Temperature observations had the highest average stockout hours at approximately **3.88**, while High Temperature observations had lower average stockout hours at approximately **3.01**. This suggests that although warmer periods may be associated with higher sales, they did not correspond to higher average stockout exposure in this analysis.

Precipitation analysis showed that higher precipitation buckets were associated with higher average sales. High Precipitation observations recorded average sales of approximately **1.14**, compared with approximately **0.98** for Low Precipitation and **1.00** for Medium Precipitation. However, the No Precipitation bucket contained a very small number of observations and had lower average sales of approximately **0.75**, so this specific comparison should be interpreted carefully.

Humidity analysis showed a similar pattern, where higher humidity levels were associated with higher average sales. Low Humidity observations recorded average sales of approximately **0.71**, Medium Humidity observations recorded average sales of approximately **1.00**, and High Humidity observations recorded average sales of approximately **1.06**. High Humidity observations also showed slightly higher stockout exposure, with average stockout hours of approximately **3.33**.

Wind level analysis showed a weaker and less consistent relationship with demand. Medium Wind observations recorded the highest average sales at approximately **1.01**, while High Wind observations recorded lower average sales at approximately **0.93**. High Wind observations also had a higher zero-sales rate of approximately **6.57%**, but overall wind level appears to be a weaker demand signal compared with temperature, precipitation, and humidity.

Overall, the weather impact analysis suggests that weather variables may be useful as contextual features in the forecasting workflow. Temperature, precipitation, and humidity show some directional relationship with sales, but the correlations are weak. Therefore, weather variables should be treated as supporting features rather than primary demand drivers. These variables may still help improve forecasting performance when combined with stronger predictors such as lag features, rolling demand features, discount variables, holiday flags, activity flags, and stockout indicators.

11.9 EDA Key Findings and Conclusion

The Exploratory Data Analysis phase provided a comprehensive understanding of historical demand behavior, product-store performance, stockout exposure, category contribution, business event impact, and weather-related demand variation.

The analysis showed that the dataset has a stable daily structure, with **50,000 product-store observations per day** across **865 products** and **898 stores**. Total sales across the training period reached approximately **4.49 million normalized sales units**, while average daily total sales were approximately **49.93K**. This confirms that the dataset contains sufficient volume and structure for time-series analysis and forecasting.

One of the strongest findings from the EDA was the importance of stockout exposure. Approximately **44.27%** of all product-store-date observations experienced at least one stockout hour, and total stockout exposure reached approximately **14.31 million hours**. Full-day stockout observations had extremely low average sales and a zero-sales rate of approximately **81.44%**, confirming that stockouts can strongly suppress observed demand.

The analysis also showed that high-demand products, stores, and categories often experience higher stockout exposure. Product-level analysis showed a positive correlation of approximately **0.72** between total sales and total stockout hours. Store-level analysis showed an even stronger correlation of approximately **0.96**, while category-level analysis also showed strong sales-stockout relationships. These findings suggest that stockout risk is not only associated with weak-performing items; it is often concentrated in high-demand areas.

Product performance analysis confirmed that demand is highly concentrated. The ABC classification showed that **121 A-class products** contributed approximately **79.86%** of total sales. This means that a relatively small group of products drives most of the business impact and should receive priority in forecasting, replenishment planning, and stockout monitoring.

Store analysis showed that certain stores function as major demand centers and also accumulate high stockout exposure. This supports the need to monitor performance not only at product level, but also at store and product-store levels.

Category analysis showed that product hierarchy is highly relevant to both demand and stockout risk. `management_group_id = 6`, `first_category_id = 4`, second-level category path 4-28, and third-level category path 4-28-81 were identified as key demand contributors. Several high-performing category

groups also showed high stockout exposure, making category-level monitoring important for business reporting and inventory planning.

Seasonality analysis showed that weekends and holidays are associated with stronger demand. Weekend observations recorded higher average sales than weekday observations, and holiday observations recorded higher average sales than non-holiday observations. These findings support the use of calendar-based features such as day of week, weekend indicator, and holiday flag during feature engineering.

Discount analysis showed that stronger discounting is associated with higher average sales. The High Discount bucket recorded average sales of approximately **2.07**, compared with approximately **0.83** for the Standard Price bucket. Activity periods were closely linked to deeper discounting, while holidays showed a clearer direct demand uplift than activity periods alone.

Weather variables showed weaker direct relationships with sales compared with product, store, category, stockout, discount, and calendar variables. Temperature, precipitation, and humidity showed some directional relationship with demand, but correlations were relatively weak. Therefore, weather variables should be treated as supporting features rather than primary demand drivers.

Overall, the EDA confirms that demand forecasting in this project should not rely on historical sales alone. Observed sales are influenced by stockout exposure, product importance, store location, category hierarchy, discounting, holidays, activity periods, and weather context. The most important insight is that low or zero sales may not always indicate weak demand; in many cases, they may reflect product unavailability.

These findings provide a solid foundation for the next stage of the project: **Feature Engineering**. The EDA results indicate that the forecasting dataset should include lag features, rolling demand features, stockout-related features, calendar variables, discount indicators, activity and holiday flags, product/category/store identifiers, and selected weather variables. This will help the forecasting model better distinguish between true demand patterns and sales values affected by stockout constraints.

12. Feature Engineering

Feature engineering was conducted to transform the prepared daily analytical dataset into a modeling-ready structure for demand forecasting. The goal of this stage was to create historical demand features, stockout-related variables, calendar indicators, business event features, and weather-based predictors that can help forecasting models learn demand patterns more effectively.

The feature engineering process was performed at the same analytical grain used throughout the project: one row represents one product-store-date observation. This ensures consistency with the earlier data preparation, data quality assessment, and EDA stages.

Calendar features were created from the date column to capture time-based demand behavior. These included day of week, weekend indicator, day of month, week of year, and month. These features were included because the EDA showed clear differences in demand between weekdays, weekends, holidays, and activity periods.

Historical sales lag features were created for each product-store time series. The generated lag features included previous-day sales, previous-week sales, two-week lag sales, and four-week lag sales. These features allow the model to learn recent demand behavior as well as weekly and longer-term demand patterns.

Rolling sales features were also created to summarize recent historical demand. These included 7-day, 14-day, and 28-day rolling sales averages, as well as 7-day rolling sales volatility. All rolling features were shifted one day before calculation to prevent data leakage. This ensures that the current day's target value is not used to create features for the same day.

Stockout-based features were added because the EDA showed that product availability has a strong effect on observed sales. These features included previous-day stockout hours, 7-day rolling average stockout hours, 7-day rolling total stockout hours, and 14-day rolling average stockout hours. These variables help the model distinguish between true low demand and constrained demand caused by product unavailability.

Business interaction features were created to capture relationships identified during the EDA. These included discount buckets, holiday-activity interaction, and weekend-holiday interaction. These features help represent combined business effects, such as promotional activity during holidays or demand changes during holiday weekends.

After feature creation, the full modeling dataset contained **4,500,000 rows and 38 columns**. Missing values appeared naturally in lag and rolling features at the beginning of each product-store time series. These missing values are expected because early observations do not have enough historical records to calculate lag and rolling features.

Rows with missing lag and rolling features were removed before model training. After this step, the final clean modeling dataset contained **3,100,000 rows and 38 columns**. This reduction is expected because the longest lag feature, `sales_lag_28`, requires 28 prior days of history for each product-store combination.

Overall, the feature engineering process produced a clean and structured training dataset that combines product, store, category, calendar, discount, weather, stockout, lag, and rolling demand features. This dataset is ready for the next stage of the project: forecasting model development, backtesting, and evaluation.

13. Forecasting Model Development

Forecasting model development was conducted to build and evaluate demand forecasting models using the feature-engineered training dataset. The objective of this stage was to determine whether historical demand patterns, product-store attributes, calendar effects, discount variables, weather conditions, and stockout-related signals could be used to predict future daily sales more accurately than simple baseline methods.

The forecasting workflow followed a time-based validation approach to simulate a real-world forecasting process. Historical observations were used for training, while the most recent period was held out for validation. This approach is important because demand forecasting should always respect chronological order, where past data is used to predict future demand.

The modeling process included four main stages: creating a time-based validation split, defining the modeling target and input features, establishing baseline forecasting models, and training a machine learning forecasting model.

13.1 Time-Based Validation Split

A time-based validation split was created from the feature-engineered modeling dataset. This approach was used instead of a random split because forecasting models must be evaluated in a way that reflects real business usage: training on historical data and validating on a later future period.

The feature-engineered modeling dataset contained **3,100,000 rows and 38 columns**, covering the period from **2024-04-25 to 2024-06-25**. The first available date in the modeling dataset is **2024-04-25** because earlier records were removed after creating lag and rolling features, especially the `sales_lag_28` feature, which requires 28 days of historical data.

The last **14 days** of the modeling dataset were selected as the validation period. The training period covered **2024-04-25 to 2024-06-11**, while the validation period covered **2024-06-12 to 2024-06-25**.

The training set contained **2,400,000 rows**, and the validation set contained **700,000 rows**. The validation size is consistent with the dataset structure because the dataset contains **50,000 product-store combinations** observed across **14 validation days**.

Both the training and validation sets contained the same business universe: **898 stores, 865 products, and 50,000 product-store pairs**. This confirms that the validation period is structurally aligned with the training period and does not introduce unseen product-store combinations.

A leakage check confirmed that the validation period starts after the training period ends. This ensures that the validation process respects chronological order and provides a reliable basis for evaluating forecasting performance.

13.2 Target, Features, and Evaluation Metrics

The forecasting target was defined as `sale_amount`, which represents the daily sales value to be predicted at the product-store-date level.

A total of **33 model input features** were selected for forecasting. These features included product identifiers, store identifiers, category hierarchy fields, calendar variables, discount variables, holiday and activity flags, weather variables, sales lag features, rolling sales features, and stockout-based features.

The training feature matrix contained **2,400,000 rows and 33 features**, while the validation feature matrix contained **700,000 rows and 33 features**. The corresponding target vectors contained the `sale_amount` values for the same observations.

Missing value validation confirmed that there were **no missing values** in the training features, validation features, training target, or validation target. This confirmed that the feature-engineered datasets were clean and ready for baseline model evaluation and machine learning model training.

Three evaluation metrics were selected to assess forecasting performance:

- **MAE — Mean Absolute Error:** measures the average absolute difference between actual and predicted sales.
- **RMSE — Root Mean Squared Error** gives more weight to larger errors and helps identify whether the model makes large forecasting mistakes.
- **WAPE — Weighted Absolute Percentage Error:** measures total absolute error as a percentage of total actual demand. This metric is especially useful in retail forecasting because it evaluates error relative to overall demand volume.

These metrics provide complementary views of model accuracy and were used to compare simple baseline forecasts with the machine learning forecasting model.

13.3 Baseline Forecasting Models

Baseline forecasting models were created to establish reference performance before training machine learning models. These models are simple forecasting approaches that help determine whether more advanced models provide meaningful additional value.

Three types of baseline forecasting methods were evaluated.

The first baseline was the **Naive Forecast**, where the forecast for each product-store-date observation was based on the previous day's sales for the same product-store combination. This was represented by the `sales_lag_1` feature.

The second baseline was the **Seasonal Naive Forecast**, where the forecast was based on sales from the same day in the previous week. This was represented by the `sales_lag_7` feature. This baseline is useful because the EDA showed that demand varies by day of week and weekend behavior.

The third baseline was the **Moving Average Forecast**, where the forecast was based on recent historical average sales. Two moving average baselines were evaluated: a 7-day rolling average and a 28-day rolling average. These were represented by `rolling_mean_7` and `rolling_mean_28`.

The best-performing baseline was the **7-day moving average forecast**, which achieved:

- **MAE:** 0.3831
- **RMSE:** 0.6312
- **WAPE:** 0.3399

The 28-day moving average baseline achieved a WAPE of approximately **0.3675**, while the naive and seasonal naive baselines achieved WAPE values of approximately **0.4124** and **0.4212**, respectively.

These results indicate that recent 7-day demand history provides a stronger forecasting benchmark than using only the previous day, the same day from the previous week, or a longer 28-day average. Therefore, the 7-day moving average forecast was selected as the main baseline benchmark for comparing machine learning models.

13.4 Machine Learning Forecasting Model

After establishing baseline performance, a machine learning forecasting model was trained using the feature-engineered dataset. The objective was to determine whether a model using multiple demand, calendar, product, store, stockout, discount, and weather signals could outperform simple historical baseline methods.

The selected model was **HistGradientBoostingRegressor**, a gradient boosting model suitable for large tabular datasets. The model was trained on **2,400,000 training rows** and evaluated on the **700,000-row validation set**.

The machine learning model achieved the following validation performance:

- **MAE:** 0.3551
- **RMSE:** 0.6193
- **WAPE:** 0.3151

This performance was better than all baseline models. Compared with the best baseline, the 7-day rolling mean forecast, the machine learning model reduced WAPE from **0.3399** to **0.3151**.

This improvement shows that the feature-engineered model provides additional forecasting value beyond simple historical averages. The model was able to use multiple signals, including lag features, rolling demand features, product and store identifiers, category hierarchy, calendar variables, discount effects, weather variables, and stockout-related indicators, to generate more accurate validation forecasts.

The validation predictions and model comparison results were saved for deeper error analysis. Overall, the machine learning model established a stronger forecasting benchmark than the baseline models and was selected for further evaluation in the next stage of the project.

13.5 Forecast Error Analysis

Forecast error analysis was conducted after model evaluation to better understand where the forecasting model performs well and where it struggles. While aggregate metrics such as MAE, RMSE, and WAPE provide an overall view of model accuracy, error analysis provides more detailed insight into model behavior across time and business dimensions.

The validation predictions showed that the model slightly underpredicted total demand during the validation period. The average actual sales value was approximately **1.1271**, while the average predicted sales value was approximately **1.0919**. Total actual validation sales reached approximately **788,935**, while total predicted sales were approximately **764,296**.

The model achieved a mean absolute error of approximately **0.3551** and a median absolute error of approximately **0.2253**. This indicates that the typical prediction error was relatively moderate, although some larger errors were present. The maximum absolute error reached approximately **21.03**, showing that a small number of product-store-date observations were more difficult to predict.

Daily error analysis showed that the model captured the overall direction of demand across the validation period, but it slightly underpredicted some high-demand days. The highest actual sales occurred on **2024-06-22** and **2024-06-23**, with total actual sales of approximately **70,919** and **70,770**, respectively. The model predicted approximately **66,883** and **67,065** for those days, which indicates that the model followed the upward pattern but underestimated the peak demand level.

Daily WAPE values ranged from approximately **29.7%** to **33.9%** across the validation period. This suggests that model performance was relatively stable day to day, although forecasting accuracy varied depending on the demand level and daily conditions.

The actual versus predicted daily sales chart showed that the model followed the validation trend reasonably well. However, the model was less accurate during sharp demand changes and peak sales days. This suggests that additional improvements may be possible through more advanced feature engineering, better promotion-related features, product-store-level segmentation, or model tuning.

Overall, the forecast error analysis confirms that the machine learning model provides useful forecasting performance and improves over the baseline methods. However, the model still has room for improvement, especially in capturing demand spikes, short-term fluctuations, and product-store-level volatility. These findings will guide the next stages of model refinement, feature importance analysis, and inventory risk segmentation.

13.6 LightGBM Base Model

After evaluating the initial machine learning model, a LightGBM model was trained to test whether a more efficient gradient boosting approach could improve forecasting performance.

The LightGBM model was trained using the same feature-engineered training and validation datasets. The model used product, store, category, calendar, discount, activity, holiday, weather, lag, rolling demand, and stockout-based features. Categorical-like fields such as product IDs, store IDs, city IDs, category IDs, day of week, and month were treated as categorical variables.

The base LightGBM model achieved a validation **MAE of 0.3459**, **RMSE of 0.5905**, and **WAPE of 0.3069**. This performance was better than both the best baseline model and the previous HistGradientBoostingRegressor model.

The best baseline model, the 7-day moving average forecast, achieved a WAPE of **0.3399**. The HistGradientBoostingRegressor model achieved a WAPE of **0.3151**. LightGBM reduced the WAPE further to **0.3069**, establishing the strongest forecasting benchmark so far.

This result indicates that LightGBM was able to capture additional demand patterns from the feature-engineered dataset, including product-store behavior, category effects, stockout history, discount signals, and calendar-based demand variation. Based on this result, LightGBM was selected for further tuning and model refinement.

13.7 Tuned LightGBM

After the base LightGBM model outperformed the previous models, a small tuning experiment was conducted to test whether different LightGBM hyperparameter settings could further improve validation performance.

Three LightGBM configurations were evaluated: a conservative model, a balanced model, and a deeper model. The objective was to reduce validation WAPE below the base LightGBM benchmark of **0.3069** while avoiding excessive training time on the large dataset.

The best-performing tuned configuration was the **balanced LightGBM model**, which achieved a validation **MAE of 0.3459**, **RMSE of 0.5933**, and **WAPE of 0.3069**.

This slightly improved the WAPE compared with the base LightGBM model, which achieved a WAPE of **0.306947**. The tuned balanced model achieved a WAPE of **0.306869**. Although the improvement was small, it represented the best WAPE among all models tested so far.

The results showed that LightGBM remained the strongest model family in this project. The base and tuned LightGBM models both outperformed the baseline forecasting methods and the initial HistGradientBoostingRegressor model. Because the tuned balanced LightGBM model achieved the best WAPE, it was selected as the leading candidate for the next stage of model comparison and ensemble testing.

13.8 Simple Ensemble Forecasting

After evaluating baseline models, HistGradientBoosting, base LightGBM, and tuned LightGBM, a simple ensemble experiment was conducted to test whether combining multiple model predictions could improve validation performance.

The ensemble approach used weighted averages of predictions from the strongest machine learning models and the 7-day moving average baseline. The objective was to determine whether combining models with different decision patterns could reduce forecasting error.

Several weighting strategies were tested. The best-performing ensemble was a weighted combination of:

- **50% Tuned LightGBM**
- **30% Base LightGBM**
- **20% HistGradientBoosting**

This ensemble achieved a validation **MAE of 0.3448**, **RMSE of 0.5918**, and **WAPE of 0.3059**.

This was the best result across all forecasting approaches tested so far. The ensemble slightly outperformed the tuned LightGBM model, which achieved a WAPE of **0.3069**, and the base LightGBM model, which achieved a WAPE of **0.3069**. It also outperformed the HistGradientBoosting model and all baseline forecasting methods.

The improvement from the ensemble was modest but meaningful, especially because the individual LightGBM models were already strong. The result suggests that combining multiple machine learning models can reduce some model-specific errors and improve overall forecast stability.

Based on validation performance, the simple ensemble became the leading forecasting candidate at this stage of the project.

13.9 Ensemble Error Analysis

After selecting the best-performing ensemble model, forecast errors were analyzed across multiple business dimensions to understand where the model performs well and where it struggles. The objective of this analysis was to determine whether further model refinement or segmented modeling would provide additional value.

The best ensemble model achieved an overall WAPE of **0.3059** on the validation period. The model slightly underpredicted total demand, with total actual sales of approximately **788,935** compared with total predicted sales of approximately **762,284**. This resulted in a prediction bias of approximately -**26,651**, indicating that the model generally forecasted below the actual validation demand.

Daily error analysis showed that the ensemble model followed the overall demand trend reasonably well. However, the model slightly underpredicted several high-demand days, particularly around **2024-06-22** and **2024-06-23**. Daily WAPE values ranged from approximately **28.5%** to **33.2%**, showing relatively stable performance across the 14-day validation period.

Product-level analysis showed that forecast errors were concentrated in several important products, including products 300, 215, 70, 834, 117, 267, and 4. Many of these products were already identified during EDA as high-sales or high-stockout products. This indicates that products with high demand volume and operational complexity remain challenging to forecast accurately.

Store-level analysis showed that some stores had larger forecast errors than others. Store 417 showed particularly high underprediction, while several high-sales stores also appeared among the highest-error stores. This suggests that store-level demand centers may require additional monitoring, especially when local demand patterns differ from the broader model behavior.

The most important result came from stockout severity analysis. The model performed relatively well for observations with no stockout, low stockout, and medium stockout. However, performance deteriorated sharply for high stockout and full day stockout observations. Full day stockout observations had extremely high WAPE because actual sales were close to zero while the model continued to predict meaningful sales. This confirms that severe stockout conditions create a major forecasting challenge.

Holiday and activity error analysis showed that the model handled holiday periods reasonably well. Holiday segments had lower WAPE than non-holiday segments, suggesting that the calendar and business event features helped the model capture holiday-related demand behavior.

ABC product class analysis showed that the model performed best for A-class products, with a WAPE of approximately **0.291**. B-class and C-class products had higher WAPE values of approximately **0.369** and **0.379**, respectively. This indicates that lower-volume products are more difficult to forecast accurately when measured by relative error, while high-impact A products benefit from stronger historical demand signals.

Category-level analysis showed that some categories contributed more absolute error than others, especially categories with higher sales volume. This suggests that category-level error monitoring may be useful for business reporting and future model refinement.

Overall, the ensemble error analysis confirmed that the model provides strong general forecasting performance but still struggles under severe stockout conditions. The most key area for improvement is stockout-aware forecasting, especially for high and full-day stockout observations. These findings support the next stage of model refinement, where stockout-aware correction or segmented modeling can be tested to reduce errors in constrained-demand situations.

13.10 Stockout-Aware Forecast Correction

After the ensemble error analysis, the largest remaining weakness was found in severe stockout conditions, especially full day stockout observations. In these cases, actual sales were often close to zero because the product was unavailable for the full monitored window. However, the model still predicted positive sales based on historical demand patterns.

To address this issue, a stockout-aware post-processing correction was tested. The objective was to reduce overprediction in high-stockout and full-day stockout cases and evaluate whether this adjustment could improve validation performance.

Several correction factors were tested for high-stockout and full-day-stockout observations. Predictions for no-stockout, low-stockout, and medium-stockout observations were kept unchanged. The best-performing correction rule reduced high-stockout predictions by 20% and set full-day stockout predictions to zero.

The original best ensemble model achieved a validation **MAE of 0.3448**, **RMSE of 0.5918**, and **WAPE of 0.3059**. After applying the stockout-aware correction, performance improved to a **MAE of 0.3200**, **RMSE of 0.5476**, and **WAPE of 0.2840**.

This was the strongest forecasting performance achieved so far in the project. The improvement confirms that severe stockout cases were a major source of forecast error and that explicitly accounting for product availability can significantly improve forecast accuracy.

The correction was especially effective for full day stockout observations. Before correction, the model strongly overpredicted sales for products that were unavailable throughout the monitored window. After correction, these predictions were reduced to zero, which substantially reduced total forecast error.

This result reinforces one of the key findings of the project: observed sales should not be interpreted independently from product availability. In fresh retail forecasting, stockout indicators are essential for distinguishing between true demand and constrained demand.

However, this correction should be interpreted carefully. The correction uses same day stockout information from the validation period. In a real forecasting scenario, actual future stockout hours may not be known at prediction time. Therefore, this approach is most appropriate when stockout information is available operationally, planned in advance, or replaced by a predicted stockout-risk signal.

Overall, the stockout-aware corrected ensemble became the best-performing forecasting approach in the validation experiment and provides a solid foundation for the final forecasting and inventory risk framework.

13.11 ABC Segment Analysis After Stockout Correction

After applying the stockout-aware correction, forecast performance was analyzed across ABC product classes to determine whether separate product-class models were needed.

The correction improved performance across all ABC classes. A-class products improved from a WAPE of **0.2906** to **0.2676**. This is the most important result because A products contributed approximately **81.05%** of total validation sales. The corrected model therefore performs strongest on the highest-impact products.

B-class products improved from a WAPE of **0.3688** to **0.3498**, while C-class products improved from **0.3790** to **0.3658**. These segments still have higher relative error than A products, which is expected because lower-volume products tend to have noisier and more intermittent demand patterns.

Although B and C products had higher WAPE values, their combined sales contribution was much smaller than A products. B products represented approximately **14.15%** of validation sales, while C products represented approximately **4.80%**. As a result, the overall business impact of forecasting error is more concentrated in A products, where the corrected model performs well.

The analysis showed that the stockout-aware correction improved all product classes, especially the most important A-class products. Therefore, a separate A/B/C model structure is not required at this stage. Instead, the global stockout-corrected ensemble model remains the strongest candidate for final model selection.

However, B and C products should continue to be monitored as potential future segmentation candidates. If future modeling work focuses on improving long-tail product forecasting, separate models or specialized intermittent-demand approaches may be tested for lower-volume products.

One important trade-off is that the corrected model increased underprediction bias across all ABC classes. This occurred because the correction intentionally reduces forecasts during severe stockout conditions. The correction improved WAPE, but it should be interpreted as an availability-aware adjustment rather than a pure unconstrained demand forecast.

Overall, the ABC segment analysis supports retaining the **global stockout-corrected ensemble** as the final forecasting approach for the current project phase.

13.12 Final Model Selection and Saving

After evaluating baseline models, HistGradientBoosting, LightGBM, tuned LightGBM, ensemble models, and stockout-aware correction, the final selected forecasting approach was the **Stockout-Corrected Ensemble Forecast**.

This approach combines three machine learning models: Tuned LightGBM, Base LightGBM, and HistGradientBoosting. The final ensemble uses a weighted average of **50% Tuned LightGBM, 30% Base LightGBM, and 20% HistGradientBoosting**.

After generating the ensemble forecast, a stockout-aware correction is applied. Forecasts for high stockout observations are reduced by **20%**, while forecasts for full day stockout observations are set to zero. This correction was introduced because the error analysis showed that severe stockout cases, especially full day stockouts, were the largest remaining source of forecasting error.

The final stockout-corrected ensemble achieved the best validation performance in the project, with a **MAE of 0.3200, RMSE of 0.5476, and WAPE of 0.2840**. This was a clear improvement over the best simple ensemble, which achieved a WAPE of **0.3059**, and the best baseline model, which achieved a WAPE of **0.3399**.

All final model components were saved for future use. These included the HistGradientBoosting model, Base LightGBM model, Tuned LightGBM model, ensemble configuration, stockout correction rules, feature metadata, and final model ranking summary. A reusable prediction function was also created to load the saved artifacts and generate forecasts without retraining the models.

The final selected approach provides the strongest validation performance and directly incorporates one of the project's core business findings: observed sales are strongly affected by product availability. Therefore, incorporating stockout-aware logic improves both forecasting accuracy and business interpretability.

14. Evaluation Forecasting

The final saved forecasting pipeline was applied to the evaluation dataset covering the period from **2024-06-26 to 2024-07-02**. The evaluation dataset contained **350,000 observations**, representing **50,000 product-store combinations** across seven future days.

To create the required lag and rolling features, the last 28 days of training data were combined with the evaluation dataset before feature engineering. This was necessary because the final model uses historical features such as previous-day sales, previous-week sales, 14-day lag, 28-day lag, and rolling demand averages.

After feature creation, only evaluation-period records were retained for forecasting. Feature validation confirmed that all required model input columns were available and that there were no missing values in the evaluation feature matrix.

The saved final model artifacts were then loaded and applied to the evaluation dataset. The final forecasting approach used the stockout-corrected ensemble, combining Tuned LightGBM, Base LightGBM, and HistGradientBoosting predictions, followed by the stockout-aware correction rule.

On the evaluation period, the model achieved a **MAE of 0.3618**, **RMSE of 0.6658**, and **WAPE of 0.3032**. This performance was slightly weaker than the validation performance, where the final stockout-corrected ensemble achieved a WAPE of approximately **0.2840**. This suggests that the evaluation period contained demand patterns that were somewhat more difficult to forecast.

Daily evaluation analysis showed that the model generally followed the direction of demand but tended to underpredict high-demand days. The largest underprediction occurred on **2024-06-30**, when actual sales reached approximately **76,012**, while predicted sales were approximately **57,164**. This indicates that the model still has difficulty fully capturing sharp demand spikes.

Overall, the evaluation results confirm that the saved final forecasting pipeline works successfully on future-period data and provides reasonable forecast accuracy. The main remaining limitation is the underprediction of sudden high-demand peaks.

15. Inventory Risk Framework

The Inventory Risk Framework was developed to translate forecasting outputs into actionable inventory and replenishment priorities. While the forecasting model predicts expected demand, the risk framework adds business context by combining forecasted demand, stockout exposure, and product importance. The objective is to identify product-store-date observations that may require operational attention and to support inventory decision-making.

The framework was built using the evaluation forecast output generated by the final stockout-corrected ensemble model. This output included actual sales, forecasted sales, stockout severity, product-store identifiers, category information, and business context variables. The framework then enriched the forecast output with ABC product classes, forecast demand tiers, stockout risk tiers, inventory risk segments, and recommended actions.

ABC product classes were added to represent product importance. A products represent the highest sales contribution, B products represent medium contribution, and C products represent lower contribution. This classification helps prioritize risk because a stockout issue affecting an A product has a greater commercial impact than a similar issue affecting a lower-contribution product.

Forecast demand tiers were created using the distribution of final predicted sales. The forecast values were divided into low, medium, high, and very high demand tiers based on forecast percentiles. This made it easier to interpret forecasted demand levels and use them in business rules. The very high forecast demand tier represented the strongest expected demand cases, while the low forecast demand tier represented lower-priority demand cases.

Stockout risk tiers were created from stockout severity. No stockout observations were classified as no availability risk, while low, medium, high, and full day stockout cases were mapped to increasing levels of availability risk. Full day stockout observations were classified as critical availability risk because the product was unavailable across the full monitored window.

The final inventory risk segmentation combined ABC class, forecast demand tier, stockout risk tier, actual sales, and forecasted sales. This produced six actionable risk segments: **Critical Risk**, **High Priority**, **Constrained Demand Candidate**, **High Demand Monitor**, **Monitor**, and **Low Priority**.

The **High Demand Monitor** segment contained the largest sales volume. It included **74,175 observations**, **117 products**, and **885 stores**, with total actual sales of approximately **219.4K** and total forecasted sales of approximately **193.8K**. These cases represent strong demand with manageable stockout exposure. The recommended action is to maintain strong product availability and monitor demand trends to prevent these cases from becoming critical.

The **Critical Risk** segment included **5,936 observations** across **93 products** and **746 stores**. This segment had average stockout hours of approximately **10.99**, indicating severe availability pressure. These cases combine important demand signals with high stockout exposure, making them the highest-priority group for replenishment review and availability monitoring.

The **Constrained Demand Candidate** segment contained **1,171 observations** across **181 products** and **606 stores**. This segment had very low actual sales of approximately **183.7**, but higher forecasted

sales of approximately **813.2** and very high average stockout hours of approximately **12.82**. This suggests that demand may be suppressed by product unavailability rather than weak customer interest. These cases are important because they highlight potential censored demand.

The **High Priority** segment included **425 observations** across **73 products** and **225 stores**, with average stockout hours of approximately **11.08**. Although smaller in volume than Critical Risk, this segment still requires priority stock review and near-term demand monitoring.

The **Monitor** segment contained **135,762 observations**, covering **816 products** and all **898 stores**. These cases represent medium-level demand or stockout exposure and should be reviewed regularly, especially if stockout exposure increases.

The **Low Priority** segment contained **132,531 observations**, covering **844 products** and all **898 stores**. These cases had lower demand and lower relative urgency, so they can be managed through standard replenishment processes.

Recommended actions were assigned to each segment to make the framework operational. Critical Risk cases require immediate replenishment review. High Priority cases require prioritized stock review. Constrained Demand Candidates require stock availability investigation. High Demand Monitor cases require continued availability monitoring. Monitor cases require regular review, while Low Priority cases can follow standard replenishment processes.

Overall, the Inventory Risk Framework connects forecasting results with operational decision-making. It helps identify which products and stores need attention, where stockout may be suppressing demand, and which product-store combinations should be prioritized in replenishment planning and dashboard monitoring.

16. Power BI Dashboard

A Power BI reporting solution was developed to transform the forecasting and inventory risk outputs into an interactive decision-support tool. Python was used to prepare the data, engineer forecasting features, train and evaluate the models, generate forecasts, and create inventory risk segments. Power BI was then used as the presentation and business intelligence layer.

The dashboard was designed for both executive and operational users. Executive users can monitor overall demand, forecasting performance, stockout exposure, and priority inventory risks. Operational users can investigate specific products, stores, dates, and recommended actions.

The Power BI solution contains four report pages:

1. Executive Overview
2. Forecast Performance
3. Inventory Risk Monitor
4. Product & Store Deep Dive

A consistent visual language was applied across the report. Actual sales were represented using dark blue, forecasted sales using teal, critical risks using red, high-priority cases using orange, constrained-demand cases using purple, and monitoring segments using blue, yellow, or neutral gray.

16.1 Executive Overview

The Executive Overview provides a concise summary of business performance, forecast accuracy, stockout exposure, and inventory priorities.

The page includes key performance indicators for:

- Total actual sales
- Total forecasted sales
- Forecast WAPE
- Forecast bias
- Critical Risk cases
- High Priority cases
- Constrained Demand Candidates
- Total stockout hours

A daily actual-versus-forecast chart shows whether the model follows the overall demand trend and highlights dates with significant forecasting gaps.

The inventory risk distribution visual shows the number of product-store-date observations in each risk segment. A separate sales exposure visual compares actual and forecasted sales across those segments. This distinction is important because a segment may contain many observations but have limited commercial impact, while another segment may contain fewer observations that require more urgent intervention.

The recommended action matrix links each risk segment to its corresponding operational response. A detailed Critical Risk table identifies the product, store, date, forecast demand, stockout severity, and action required for the highest-priority cases.

Overall, the page answers four executive questions:

- What happened?
 - What was forecasted?
 - Where is the inventory risk?
 - What action is required?
-

16.2 Forecast Performance

The Forecast Performance page evaluates the quality and behavior of the final forecasting approach.

The page includes:

- Actual sales
- Adjusted forecast sales
- WAPE
- MAE
- RMSE
- Forecast bias

The daily actual-versus-forecast trend shows how well the model follows changes in demand over the seven-day evaluation period. Daily WAPE and bias visuals identify the dates on which the model performed well and the dates on which it underpredicted or overpredicted demand.

The evaluation showed that the model generally followed the direction of daily sales but had difficulty capturing sharp demand increases. The most important example occurred on **30 June 2024**, when actual sales reached approximately **76.0K**, compared with an adjusted forecast of approximately **57.2K**.

Product-level and store-level error visuals identify the entities contributing the most to total forecast error. An additional stockout-severity visual shows how forecast performance changes under different availability conditions.

This page helps users distinguish between:

- Normal forecasting variation
 - Systematic underprediction or overprediction
 - Product-specific errors
 - Store-specific errors
 - Errors related to stockout conditions
-

16.3 Inventory Risk Monitor

The Inventory Risk Monitor converts forecasts and availability signals into operational inventory priorities.

The page focuses on six risk segments:

- Critical Risk
- High Priority
- Constrained Demand Candidate
- High Demand Monitor
- Monitor
- Low Priority

The primary indicators show the number of Critical Risk, High Priority, and Constrained Demand cases, together with the total stockout exposure and the overall share of at-risk observations.

The inventory risk distribution visual shows the volume of cases in each segment, while the stockout exposure visual shows the total availability pressure associated with those segments.

ABC product classification is incorporated to distinguish commercially important products from lower-contribution products. A Critical Risk case involving an A-class product receives greater operational priority because A products contribute the majority of total sales.

The recommended-action matrix translates analytical segments into clear operational responses.

Examples include:

- Immediate replenishment review
- Priority stock review
- Investigation of potentially suppressed demand
- Maintenance of strong availability
- Regular monitoring
- Standard replenishment

A priority product-store table provides the detailed records required for operational follow-up.

The page also includes an analysis of potential demand suppression. These cases have low realized sales, a higher pre-correction demand signal, and severe stockout exposure. The analysis helps identify circumstances where low sales may reflect product unavailability rather than genuinely weak demand.

16.4 Product & Store Deep Dive

The Product & Store Deep Dive supports detailed investigation of individual products, stores, cities, ABC classes, and dates.

Users can select a product or store and review:

- Actual sales
- Adjusted forecast sales
- Forecast WAPE
- Forecast bias
- Stockout rate
- Critical Risk cases
- Daily actual-versus-forecast behavior
- Risk segment distribution
- Product performance across stores
- Store performance across products
- Stockout severity
- Recommended actions

The page supports three main analytical scenarios.

Product Analysis

Users can select a product and compare its demand, forecast accuracy, stockout exposure, and risk distribution across stores.

Store Analysis

Users can select a store and identify the products responsible for the largest sales contribution, forecast error, or inventory availability risk.

Product-Store Analysis

Users can select both a product and a store to investigate daily performance at the most granular operational level.

This page connects executive indicators with the underlying product-store-date records and allows business users to move from a high-level risk signal to a specific operational case.

17. Key Insights

The project generated several important findings across demand behavior, forecasting performance, stockout exposure, and inventory prioritization.

17.1 Demand Is Highly Concentrated

ABC analysis showed that **121 A-class products contributed approximately 79.86% of total training sales**. In comparison, 167 B-class products contributed approximately 15.10%, while 577 C-class products contributed approximately 5.04%.

This concentration indicates that improvements in forecasting and availability for a relatively small number of products can have a substantial commercial impact.

17.2 High Demand and Stockout Exposure Are Closely Connected

Products, stores, and categories with higher demand frequently accumulated more stockout hours.

The correlation between product sales and total product stockout hours was approximately **0.72**, while the store-level relationship was approximately **0.96**. Strong relationships were also observed across product-category levels.

This suggests that stockout exposure is often the result of demand pressure rather than weak product performance.

17.3 Full-Day Stockout Strongly Suppresses Observed Sales

Full-day stockout observations recorded sales close to zero and had a zero-sales rate of approximately **81.44%** during exploratory analysis.

This creates a censored-demand problem. Recorded sales may represent what the business was able to sell, rather than what customers were willing to purchase.

Therefore, low sales should not automatically be interpreted as low demand when availability was severely constrained.

17.4 Weekends, Holidays, and Discounts Influence Demand

Weekend and holiday observations recorded higher average sales than regular weekdays and non-holiday periods.

Stronger discounts were also associated with higher average demand and greater stockout exposure. Promotional decisions should therefore be coordinated with inventory availability to prevent demand stimulation from creating additional lost-sales risk.

17.5 Machine Learning Outperformed Simple Baselines

The best simple baseline was the seven-day moving average, with validation WAPE of approximately **33.99%**.

HistGradientBoosting improved WAPE to approximately **31.51%**, while the base and tuned LightGBM models reduced WAPE to approximately **30.69%**.

A weighted ensemble of Tuned LightGBM, Base LightGBM, and HistGradientBoosting achieved WAPE of approximately **30.59%** before the availability adjustment.

17.6 Availability-Aware Adjustment Improved Validation Performance

The error analysis showed that severe stockout cases were a major source of forecasting error.

Reducing High Stockout predictions by 20% and setting Full-Day Stockout predictions to zero improved validation performance to:

- MAE: **0.3200**
- RMSE: **0.5476**
- WAPE: **28.40%**

This confirms that availability information is highly relevant when the objective is to predict realized sales.

17.7 Evaluation Performance Was Reasonable but Weaker Than Validation

On the separate seven-day evaluation period, the final saved pipeline achieved:

- MAE: **0.3618**
- RMSE: **0.6658**
- WAPE: **30.32%**

The model followed the general demand direction but underpredicted several high-demand days. The largest gap occurred on 30 June 2024.

17.8 The Model Performed Best on A-Class Products

After stockout-aware correction, WAPE by ABC class was approximately:

- A products: **26.76%**
- B products: **34.98%**
- C products: **36.58%**

A products have higher demand volume and more stable historical patterns, which makes them easier to forecast. B and C products are more intermittent and noisy.

17.9 Inventory Risk Is Concentrated in a Manageable Number of Cases

The Inventory Risk Framework identified:

- **5,936 Critical Risk observations**
- **425 High Priority observations**
- **1,171 Constrained Demand Candidates**
- **74,175 High Demand Monitor observations**

Critical cases represented a small proportion of all observations but were distributed across 93 products and 746 stores. This confirms that targeted prioritization is more practical than applying a uniform inventory policy across the entire network.

18. Business Recommendations

18.1 Prioritize A-Class Products

A-class products should receive the highest forecasting, replenishment, and availability-monitoring priority because they generate the majority of sales.

Recommended actions include:

- More frequent demand monitoring
 - Tighter replenishment thresholds
 - Higher service-level targets
 - Faster escalation of stockout cases
 - Separate review of A products classified as Critical Risk
-

18.2 Establish a Critical Risk Replenishment Queue

Critical Risk cases should be presented as an operational work queue rather than only a dashboard metric.

The queue should include:

- Product
- Store
- Date
- ABC class
- Expected demand
- Stockout severity
- Recommended action

The highest-risk product-store combinations should be reviewed first during replenishment planning.

18.3 Protect High Demand Monitor Products

High Demand Monitor contained the largest sales volume. These products are not currently in severe availability distress, but they represent substantial commercial value.

The business should protect this segment through:

- Availability threshold alerts
- Monitoring of recent stockout movement
- Promotion-aware replenishment
- Store-level safety-stock review

The objective is to prevent these products from moving into Critical Risk.

18.4 Investigate Constrained Demand Candidates

Constrained Demand Candidates should not be treated as ordinary low-sales products.

The business should investigate:

- Whether the product was unavailable during peak demand hours
- Whether replenishment arrived too late
- Whether observed sales are materially below the demand signal
- Whether the issue is isolated to specific stores
- Whether repeated stockouts indicate a permanent replenishment problem

These cases may represent potential lost-sales opportunities.

18.5 Coordinate Promotions with Inventory Availability

High discounts were associated with stronger sales and higher stockout pressure.

Before launching a major promotion, the business should review:

- Expected demand uplift
- Current availability
- Historical promotion performance
- Store-level stockout risk
- Replenishment lead time

Promotions should not be evaluated using sales uplift alone. Lost-sales exposure and product availability should also be included.

18.6 Use Segment-Specific Inventory Policies

Different product groups require different inventory strategies.

Segment	Suggested approach
A Products	High service level and frequent monitoring
B Products	Regular review and balanced inventory control
C Products	Simplified replenishment and exception-based management
Critical Risk	Immediate operational review
High Demand Monitor	Preventive availability protection
Constrained Demand	Availability and suppressed-demand investigation
Low Priority	Standard replenishment process

18.7 Monitor Forecast Bias and Demand Peaks

The model showed a tendency toward underprediction during high-demand periods.

The business should introduce:

- Alerts for unusually strong demand uplift
 - Peak-demand override rules
 - Manual review for major holidays and campaigns
 - Rolling monitoring of daily forecast bias
 - Periodic model recalibration
-

18.8 Use Power BI as an Operational Monitoring Layer

The dashboard should be refreshed regularly and used during:

- Replenishment review meetings
- Promotion planning
- Store performance monitoring
- Category management
- Forecast performance evaluation

The focus should remain on exceptions and priority actions, not only aggregate performance.

19. Limitations

19.1 Short Historical Window

The training data covered approximately 90 days. This period is useful for short-term demand forecasting but insufficient for learning annual seasonality or long-term structural changes.

19.2 Seven-Day Final Evaluation Period

The final evaluation covered only seven days, from 26 June to 2 July 2024.

Although the evaluation contained 350,000 observations, the short time horizon limits the ability to assess performance across multiple seasons, holidays, and operational environments.

19.3 Single Time-Based Validation Split

Model selection was based on one 14-day time-based validation period.

A single holdout does not provide the same robustness as rolling-origin backtesting across multiple historical windows.

19.4 Stockout Correction Uses Same-Day Availability Information

The stockout-aware adjustment used `stock_hour6_22_cnt` from the same forecast date.

This produced a strong estimate of availability-adjusted realized sales, but actual future stockout hours may not be known at the time the forecast is generated.

Therefore, the corrected forecast should not be interpreted as a deployable unconstrained demand forecast unless future availability is known, planned, or separately predicted.

19.5 Realized Sales Are Not Equal to Unconstrained Demand

The project predicts observed sales, which may already be limited by product availability.

The model does not fully reconstruct demand that would have occurred if every product had remained available throughout the day.

19.6 Limited Inventory Variables

The dataset did not include direct inventory quantities, reorder points, delivery lead times, purchase orders, safety stock, shelf capacity, or holding costs.

The Inventory Risk Framework therefore uses stockout hours as an availability-risk proxy rather than calculating true inventory sufficiency.

19.7 Limited Product and Store Descriptions

Product, store, city, and category fields were represented primarily by numeric identifiers. The absence of descriptive business names limits the immediate interpretability of the Power BI report.

19.8 Long-Tail Products Are Harder to Forecast

B and C products recorded higher relative errors because lower volume demand is often intermittent, sparse, and more sensitive to individual transactions.

19.9 Risk Thresholds Are Analytical Rules

Forecast demand tiers were based on percentile thresholds, while inventory risk segments were created using business rules.

These thresholds provide a practical prioritization framework, but they have not yet been financially optimized using lost-sales costs, holding costs, or service-level targets.

20. Future Work

20.1 Rolling Back testing

Future model evaluation should use multiple rolling validation windows rather than a single holdout.

This would provide more reliable estimates of:

- Average forecasting performance
 - Variability across weeks
 - Stability during holidays and promotions
 - Sensitivity to changing demand conditions
-

20.2 Separate Demand and Availability Models

A stronger production framework should separate:

1. Unconstrained demand forecasting
2. Stockout or availability-risk prediction
3. Realized sales estimation

This would prevent same-day actual stockout information from being required during future forecasting.

20.3 Add Inventory and Supply-Chain Variables

Future data collection should include:

- On-hand inventory
- Reorder points
- Safety stock
- Delivery lead time
- Purchase orders
- Supplier reliability
- Shelf capacity
- Replenishment frequency
- Expiration or spoilage information

These variables would allow the project to move from risk classification to inventory optimization.

20.4 Improve Peak-Demand Forecasting

Additional features should be developed for sudden demand increases, including:

- Promotion intensity
 - Days before and after holidays
 - Pay-cycle effects
 - Event countdown features
 - Interactions between discount, holiday, and weekend
 - Recent demand acceleration
 - Store-category promotion history
-

20.5 Explore Intermittent-Demand Models

Specialized approaches should be tested for B and C products, including:

- Croston-style methods
- TSB methods
- Zero-inflated models
- Separate occurrence and quantity models
- Segment-specific machine learning models

These approaches may improve forecasting for low-volume and intermittent products.

20.6 Add Prediction Intervals

Future forecasts should include uncertainty ranges:

- 1 Lower Forecast
- 2 Expected Forecast
- 3 Upper Forecast

This would help inventory teams make decisions according to different service levels and risk tolerances.

20.7 Financially Optimize Risk Segments

Risk thresholds should eventually incorporate:

- Estimated lost-sales value
- Inventory holding cost
- Replenishment cost
- Product margin
- Spoilage risk
- Service-level targets

This would convert the current analytical prioritization into an economically optimized framework.

20.8 Automate the Pipeline

The project can be extended into an automated workflow that:

1. Loads of new daily data
 2. Validates data quality
 3. Creates forecasting features
 4. Loads saved models
 5. Generates predictions
 6. Applies availability logic
 7. Refreshes inventory risk segments
 8. Updates the Power BI dataset
-

20.9 Enhance Power BI Deployment

Future Power BI improvements may include:

- Scheduled refresh
- Row-level security
- Drill-through pages
- Custom report tooltips
- Alert thresholds
- Mobile layouts
- Bookmarks and navigation
- Exportable replenishment lists

Appendix A. Repository Structure

The repository should separate code, data, outputs, models, dashboards, and documentation

FreshRetailNet-50K/

```
|
|
├── data/
|   ├── 00_raw/
|   ├── 01_prepared/
|   ├── 02_validation/
|   └── 03_modeling/
|
|
├── notebooks/
|   ├── 01_data_preparation.ipynb
|   ├── 02_data_quality.ipynb
|   ├── 03_eda.ipynb
|   ├── 04_feature_engineering.ipynb
|   ├── 05_forecasting.ipynb
|   └── 06_inventory_risk.ipynb
|
|
├── scripts/
|   ├── feature_engineering.py
|   ├── forecasting.py
|   └── inventory_risk.py
|
|
├── models/
|   ├── hist_gradient_boosting_model.joblib
|   ├── lightgbm_base_model.joblib
|   ├── lightgbm_tuned_balanced_model.joblib
|   └── final_model_config.json
```

- | └─ model_metadata.json
- |
- | └─ outputs/
- | | └─ 01_eda/
- | | └─ 02_forecasting/
- | | └─ 03_evaluation/
- | | └─ 04_inventory_risk/
- | | └─ 05_power_bi/
- |
- | └─ power_bi/
- | | └─ FreshRetail_Demand_Inventory_Intelligence.pbix
- |
- | └─ reports/
- | | └─ final_project_report.pdf
- |
- | └─ README.md
- | └─ LICENSE
- └─ .gitignore

Appendix B. License Scope Notice

License Scope Notice

The MIT License included in this repository applies only to the original source code, scripts, notebooks, analytical logic, and project documentation created for this project.

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Generated model artifacts, forecast files, and dashboard outputs are provided for educational and portfolio demonstration purposes. These outputs should not be interpreted as verified commercial forecasts or production inventory recommendations without additional validation.